

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-1: Digital Image Fundamentals

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-1&2)

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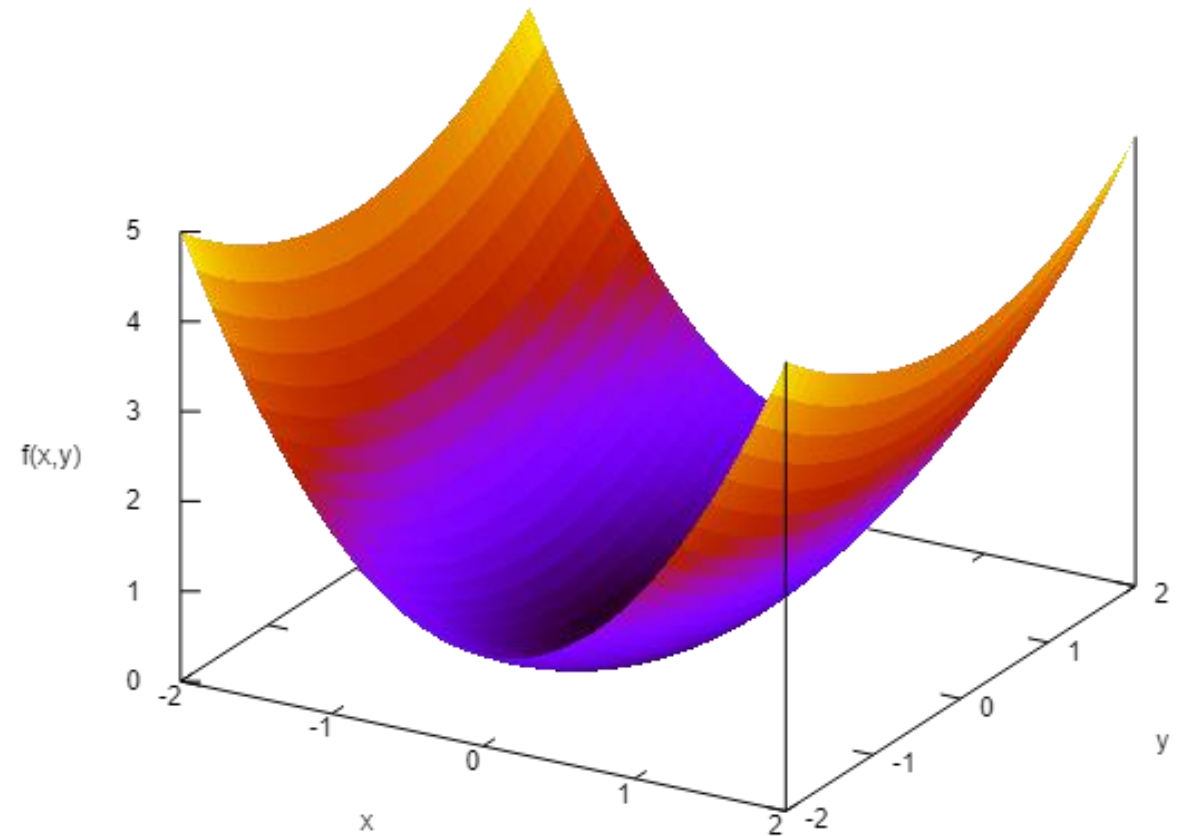
1. Image

An image can be described as a two-dimensional function, $f(x,y)$

where:

x and y represent spatial (plane) coordinates,

$f(x,y)$ is the amplitude (intensity or gray level) of the image at the point (x,y) .



1.1 Digital Image

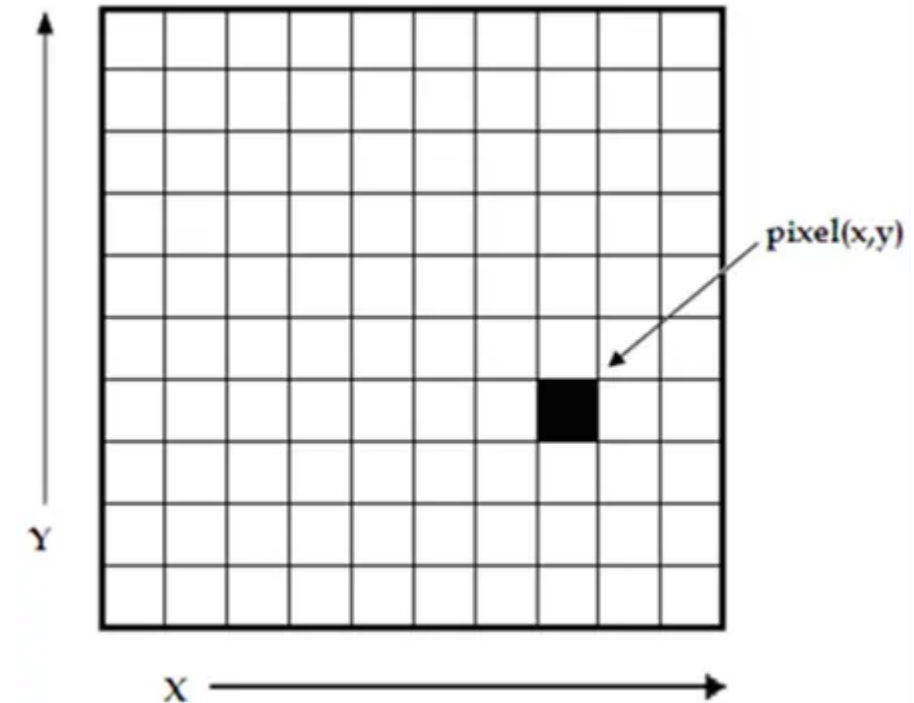
When x , y , and the intensity values of f are finite and discrete, the image is called a digital image.

1.2 Digital Image Processing

It refers to processing digital images using a digital computer.

1.3 Pixels

A pixel is the smallest unit of a digital image, representing the intensity or color at a specific location in the image.



2. Application of digital image processing

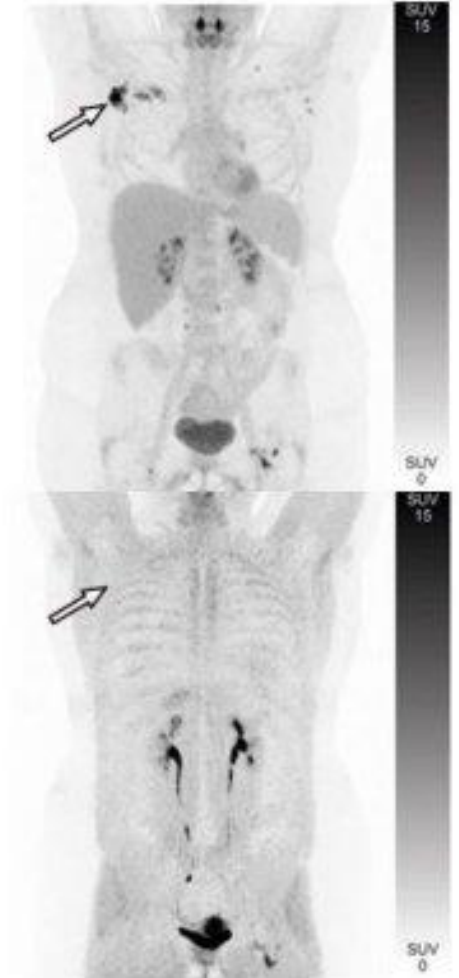
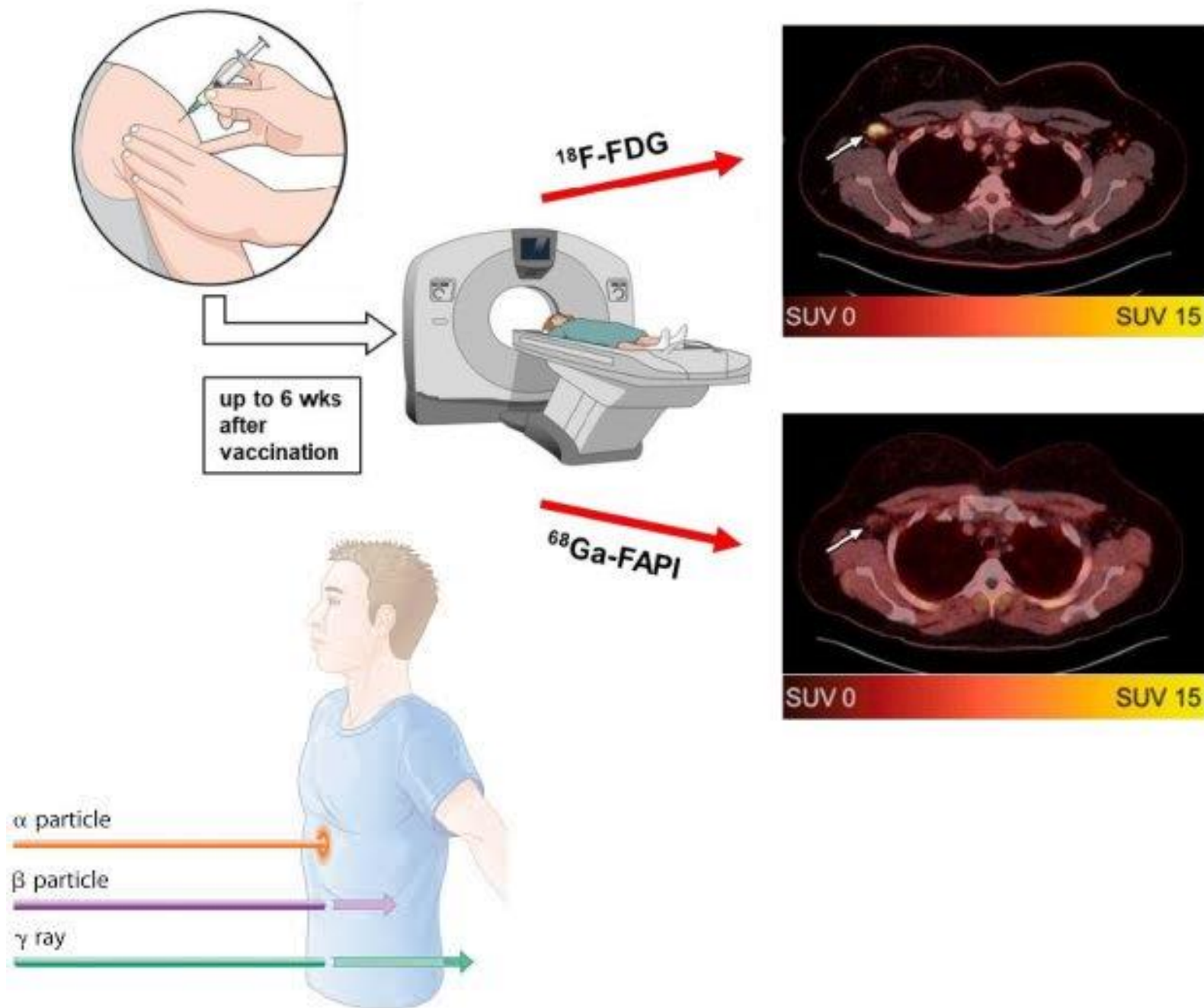
2.1 Gamma-Ray Imaging

2.1.1 Nuclear Medicine:

The radiotracer is injected into the patient's body, where it accumulates in specific tissues or organs emit gamma rays as they decay.

Gamma ray detectors, such as gamma cameras, detect these emissions, creating images of the radiotracer's distribution.

This imaging helps diagnose or monitor conditions like cancer, heart disease, or bone abnormalities.



2.1.2 Positron Emission Tomography (PET)

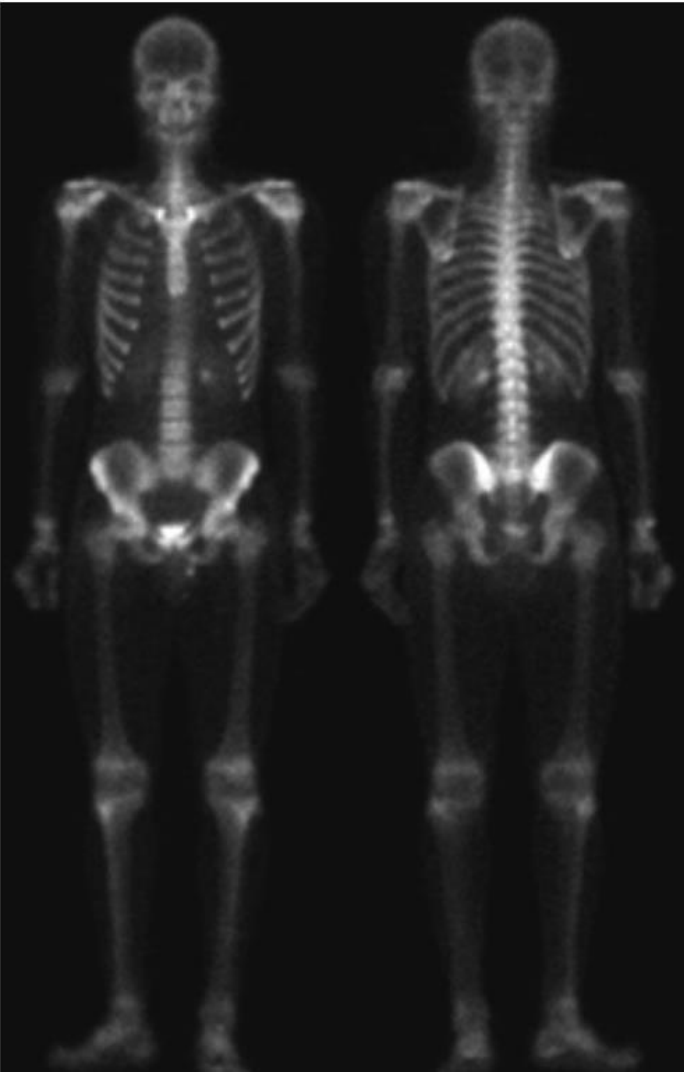
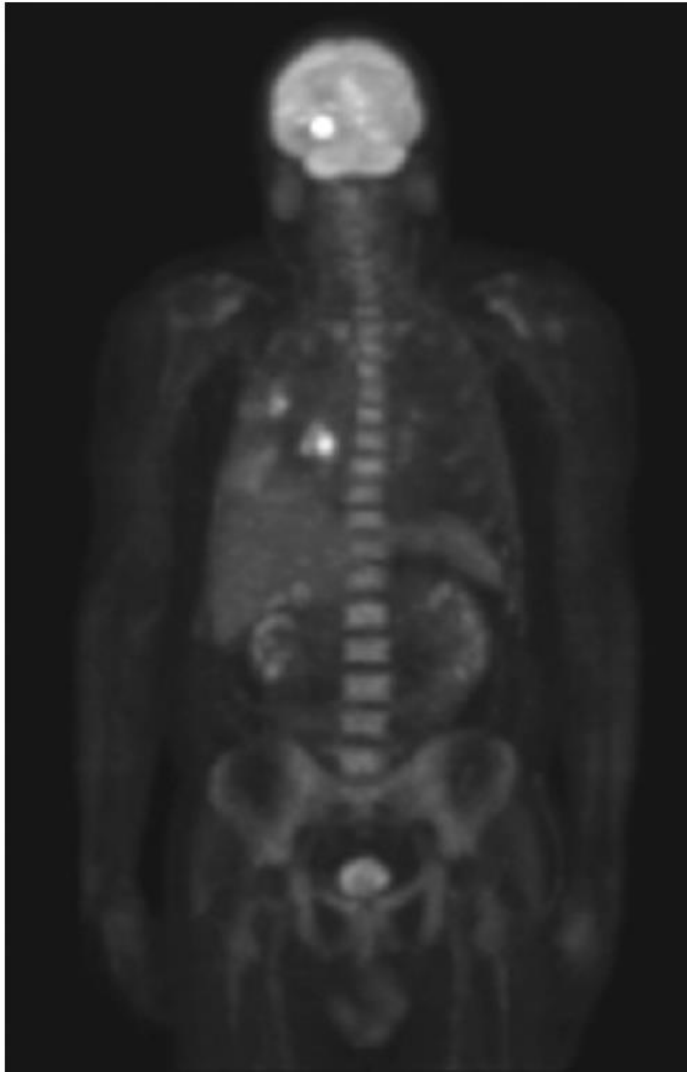
A radioactive tracer (commonly a compound like glucose labeled with a positron-emitting isotope) is injected into the patient.

The tracer travels through the body and accumulates in specific tissues, especially those with high metabolic activity, like cancer cells.

The radioactive tracer emits positrons (positively charged particles). When a positron meets an electron in the body, they annihilate each other, producing two gamma rays that travel in opposite directions.

A PET scanner detects these gamma rays.

Positron Emission Tomography (PET)



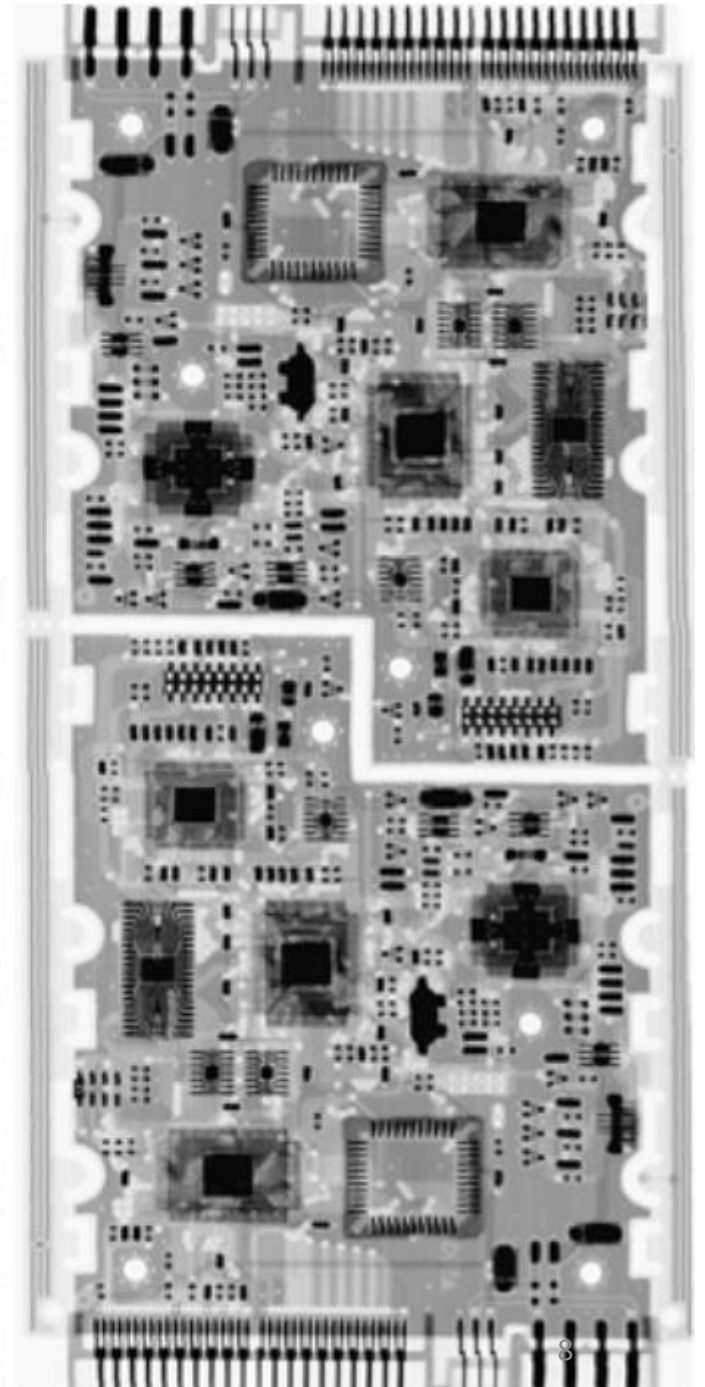
Gamma-ray Imaging

2.2 X-ray Imaging

X-rays pass through the body,
with different tissues absorbing
varying amounts of radiation.

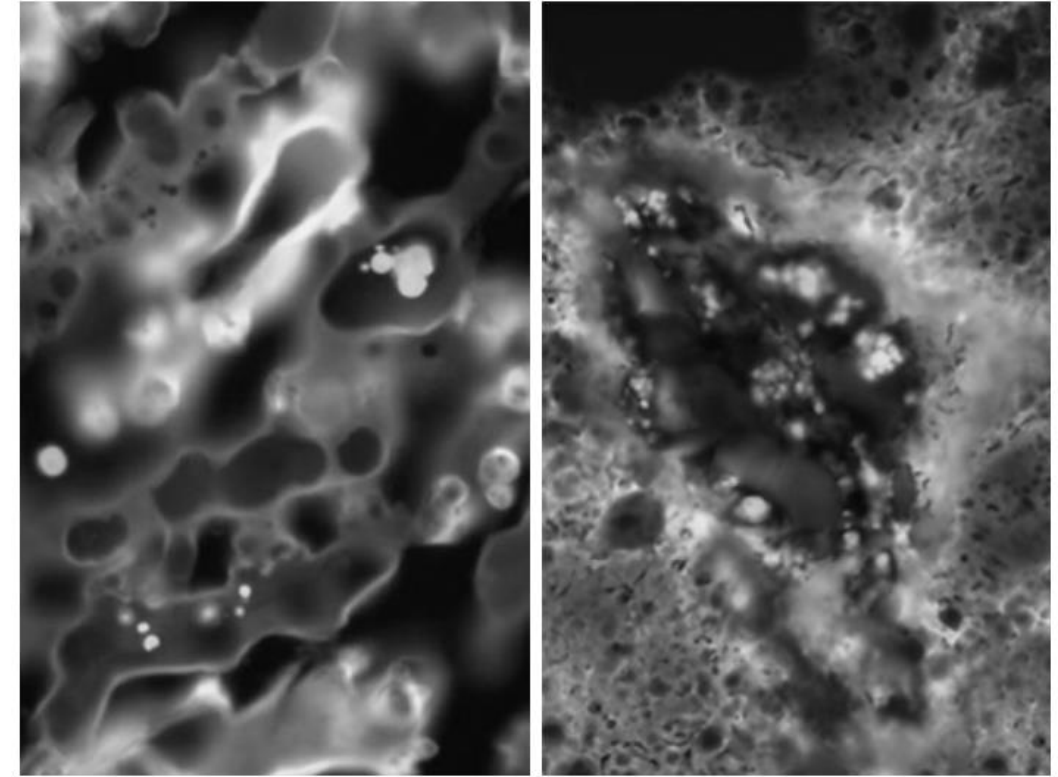
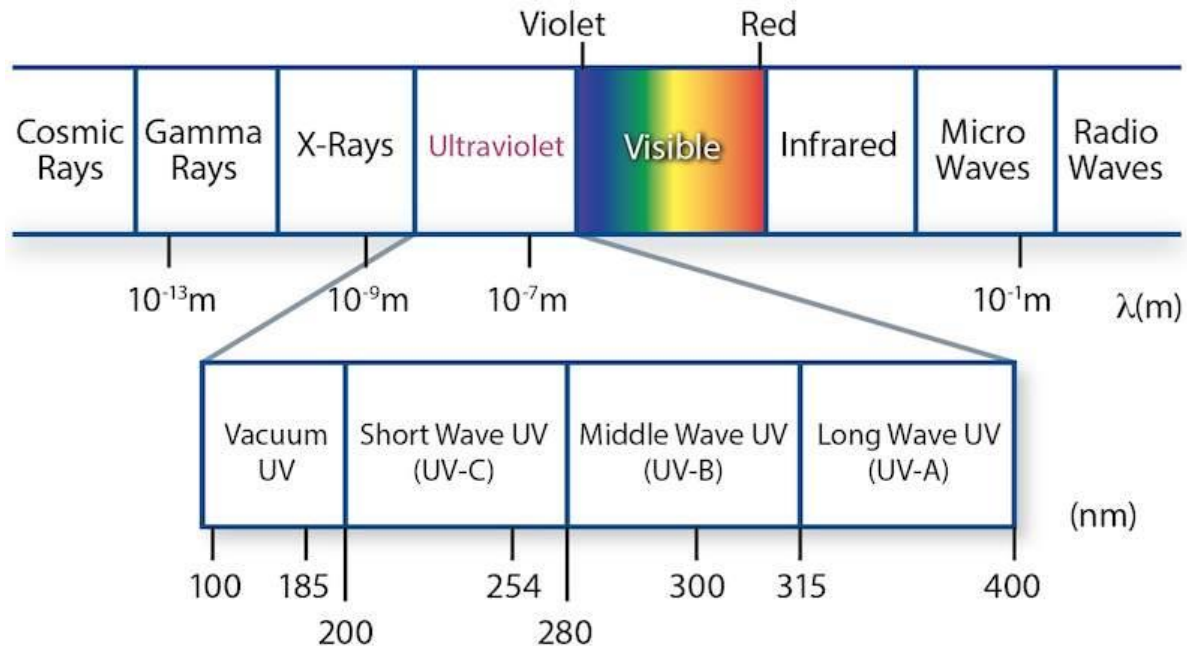
Dense structures like bones
absorb more X-rays and appear
white on the image.

Softer tissues absorb fewer X-
rays and appear in shades of
gray, while air-filled areas
appear black.



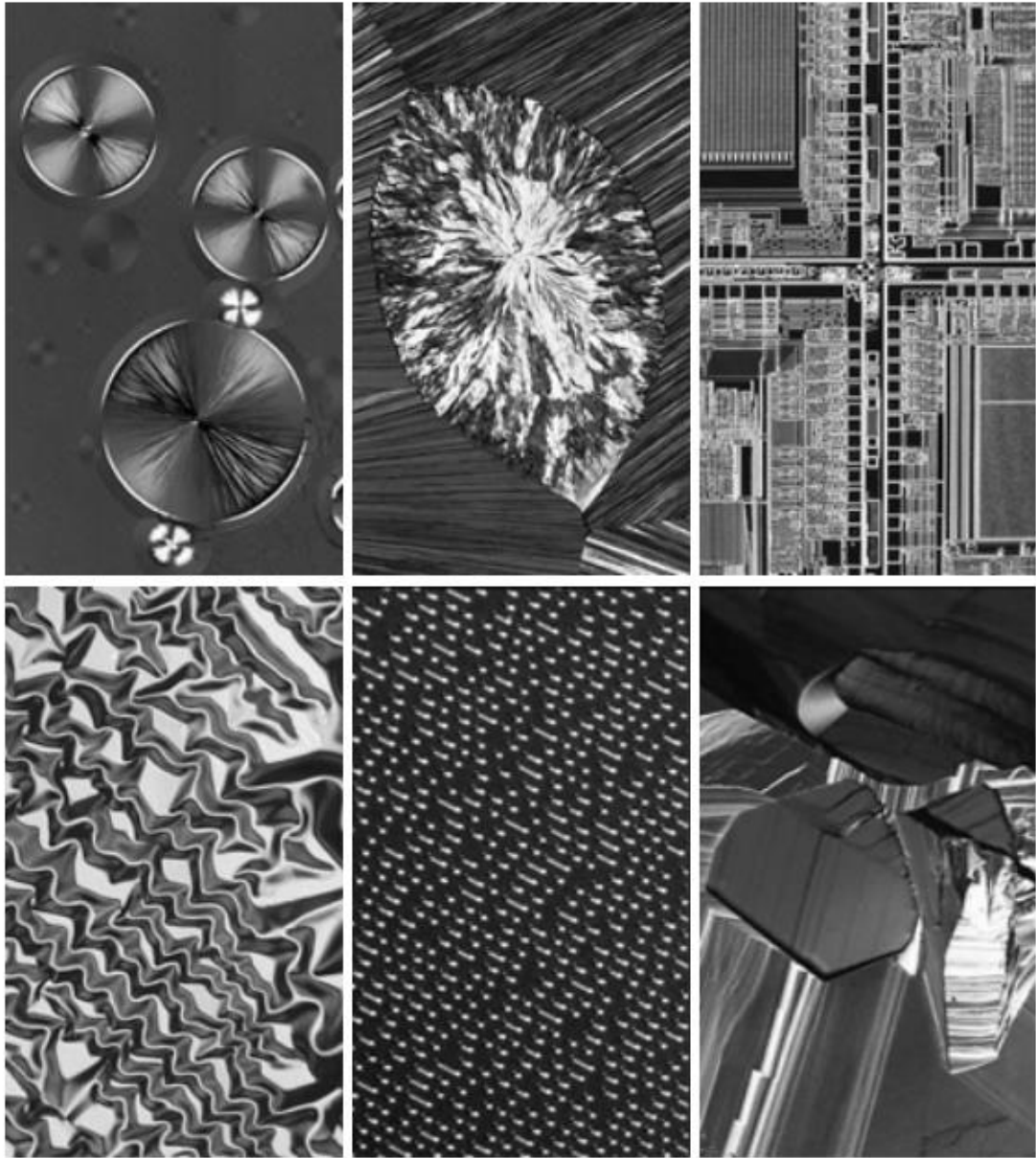
2.3 Imaging in the Ultraviolet (UV) Band

Imaging in the Ultraviolet (UV) Band involves *capturing light in the ultraviolet spectrum (wavelengths approximately 10–400 nanometers)*, which is *shorter than visible light but longer than X-rays*.



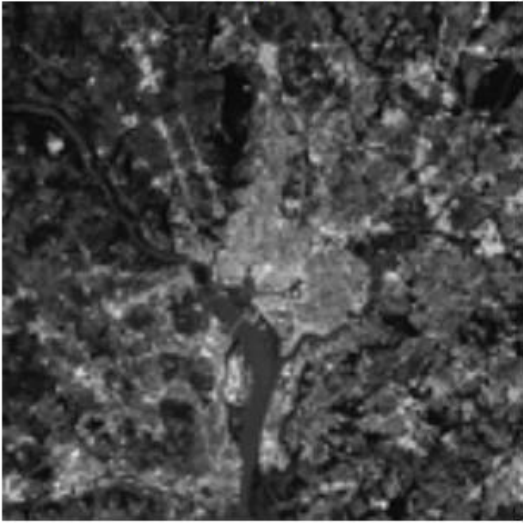
2.4 Imaging in the Visible and Infrared Bands

Examples of light microscopy images

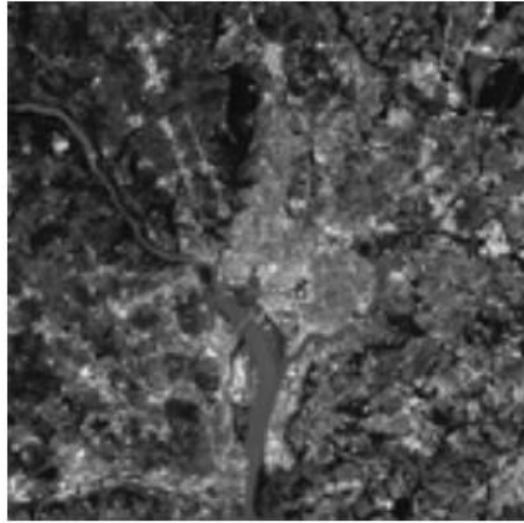


2.5 LANDSAT Satellite Images

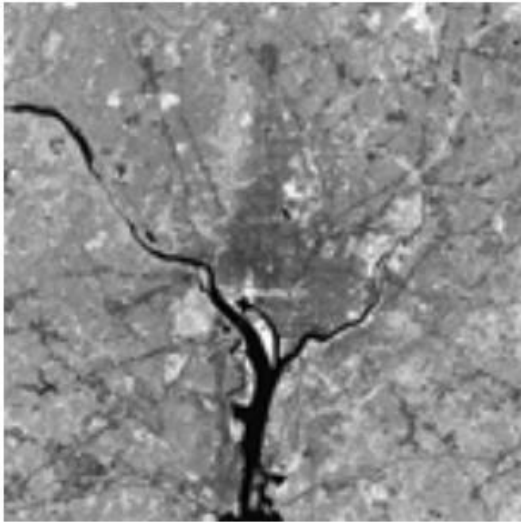
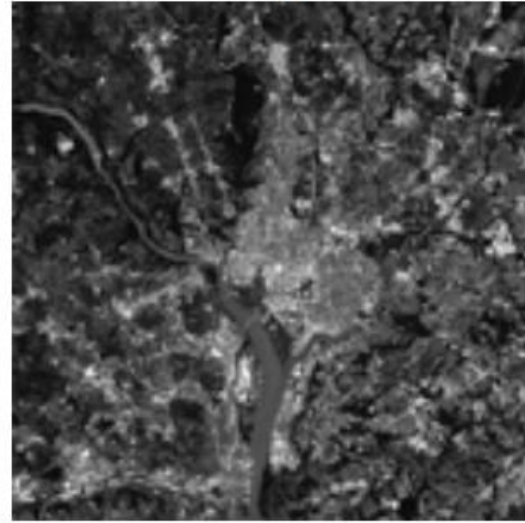
1



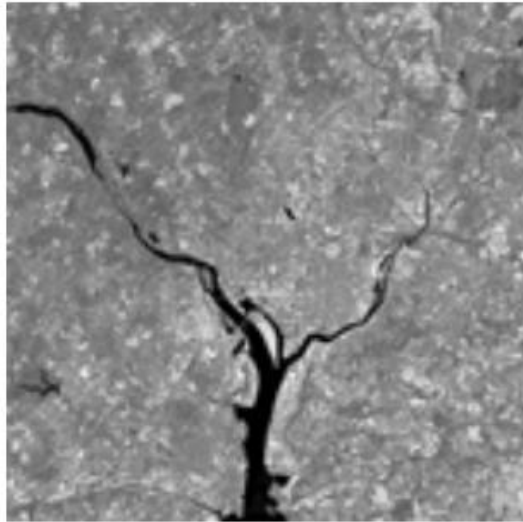
2



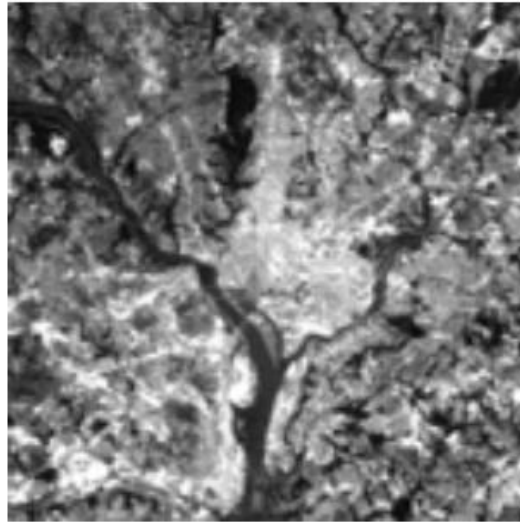
3



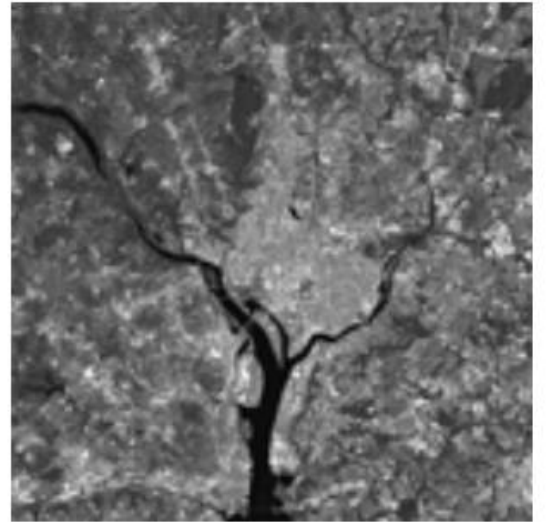
4



5

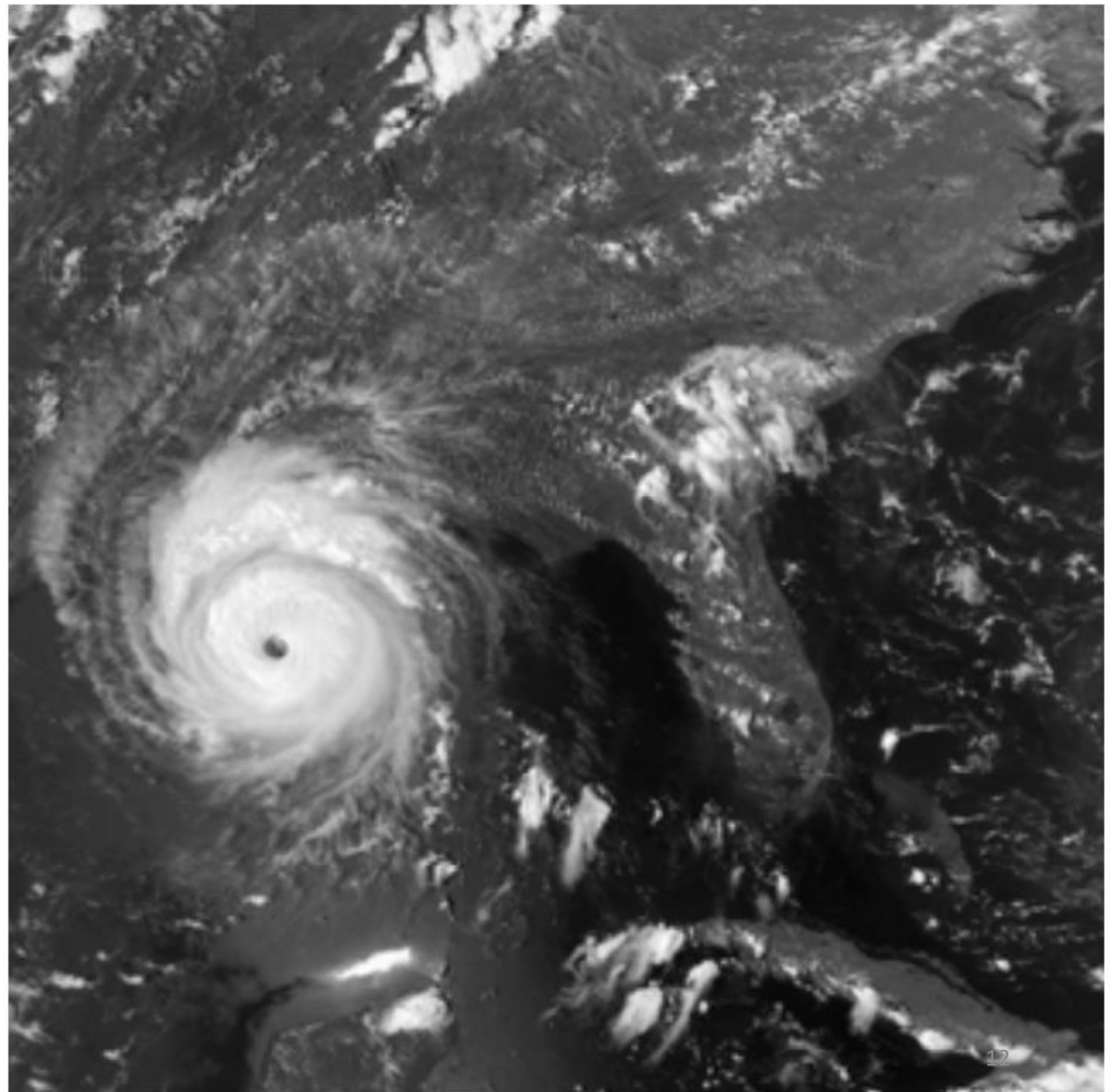


6



7

Multispectral image of Hurricane Andrew taken by NOAA GEOS (Geostationary Environmental Operational Satellite) sensors.



2.6 Infrared satellite images



2.7 Imaging in the Microwave Band

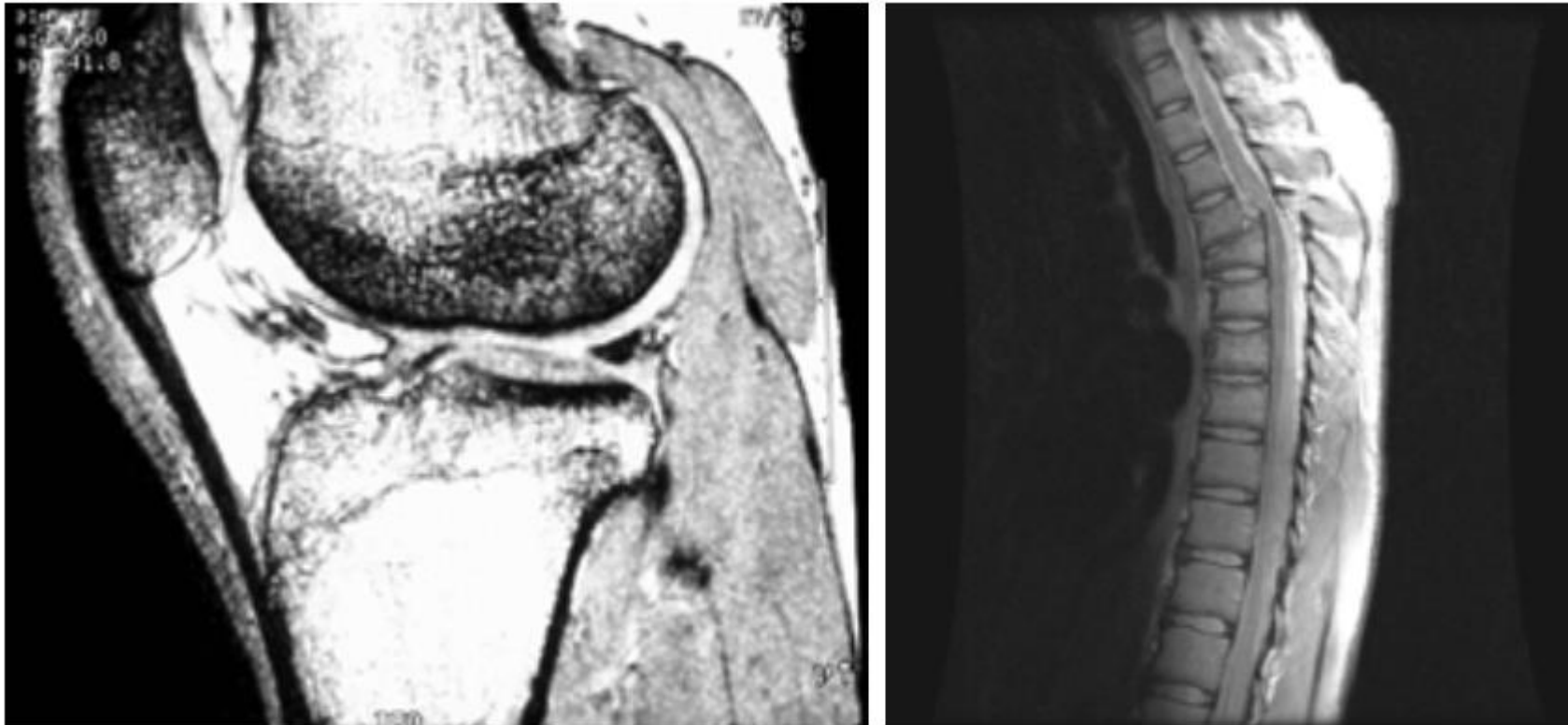
Imaging in the Microwave Band involves the use of electromagnetic waves in the microwave spectrum (wavelengths ranging from 1 millimeter to 1 meter) to capture information about objects or environments.

This technique is particularly useful due to its ability to penetrate clouds, fog, and even soil, providing insights that are inaccessible with visible or infrared imaging.



2.8 Magnetic Resonance Imaging (MRI)

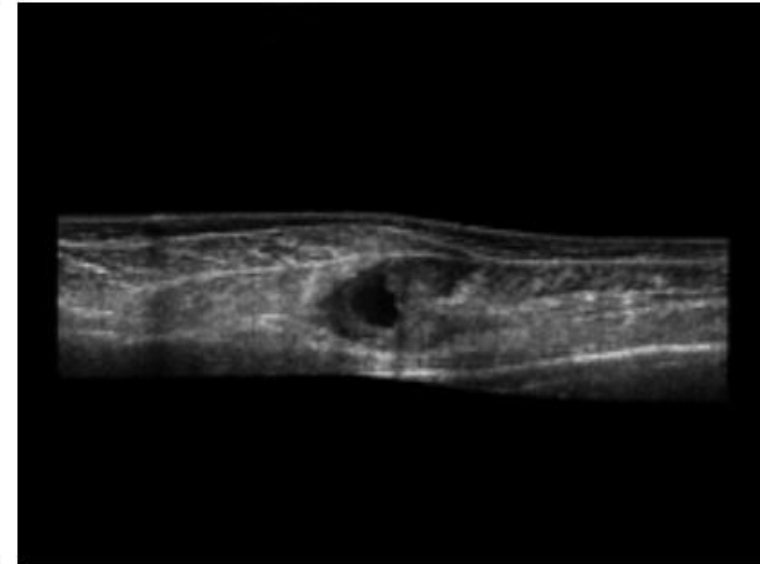
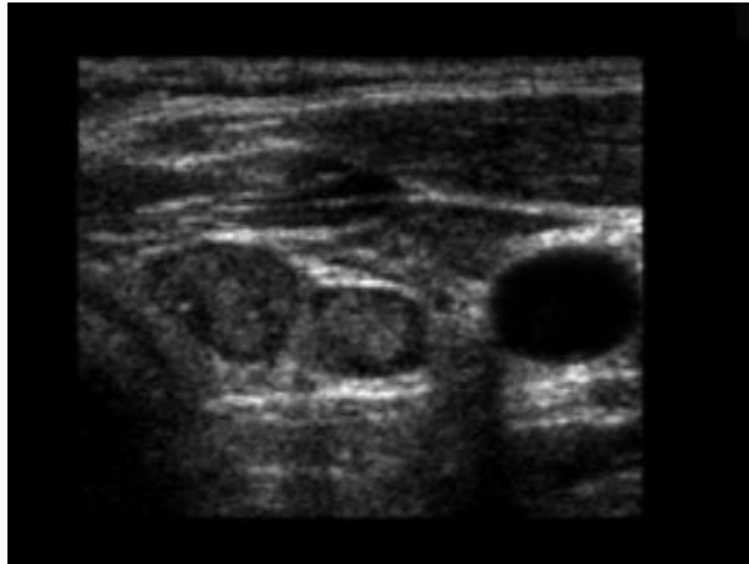
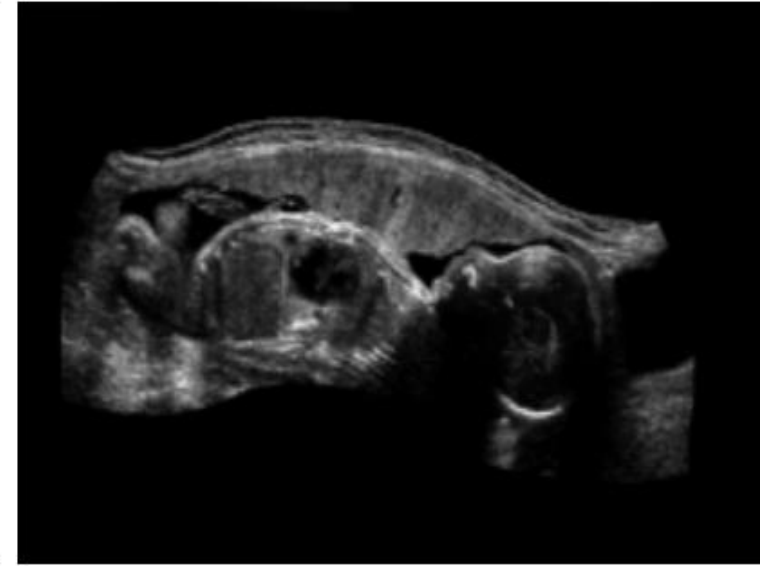
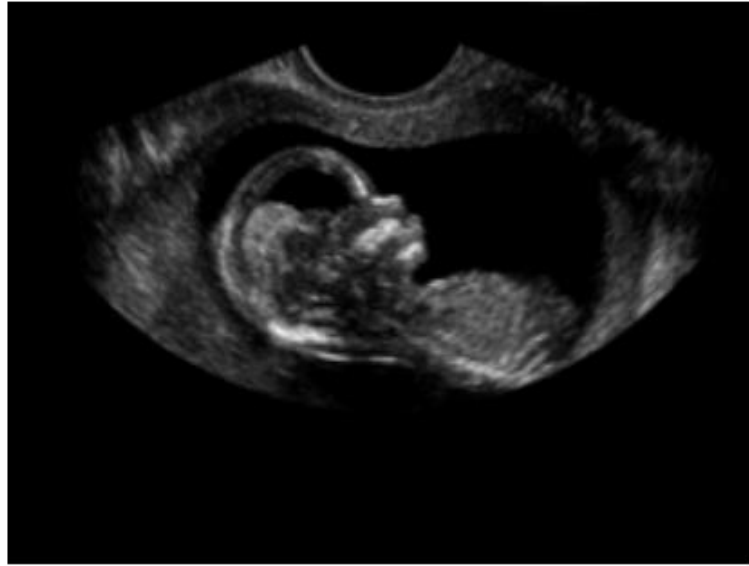
Magnetic Resonance Imaging (MRI) is a medical imaging technique that uses strong magnetic fields and radio waves to generate detailed images of the body's internal structures, particularly soft tissues.



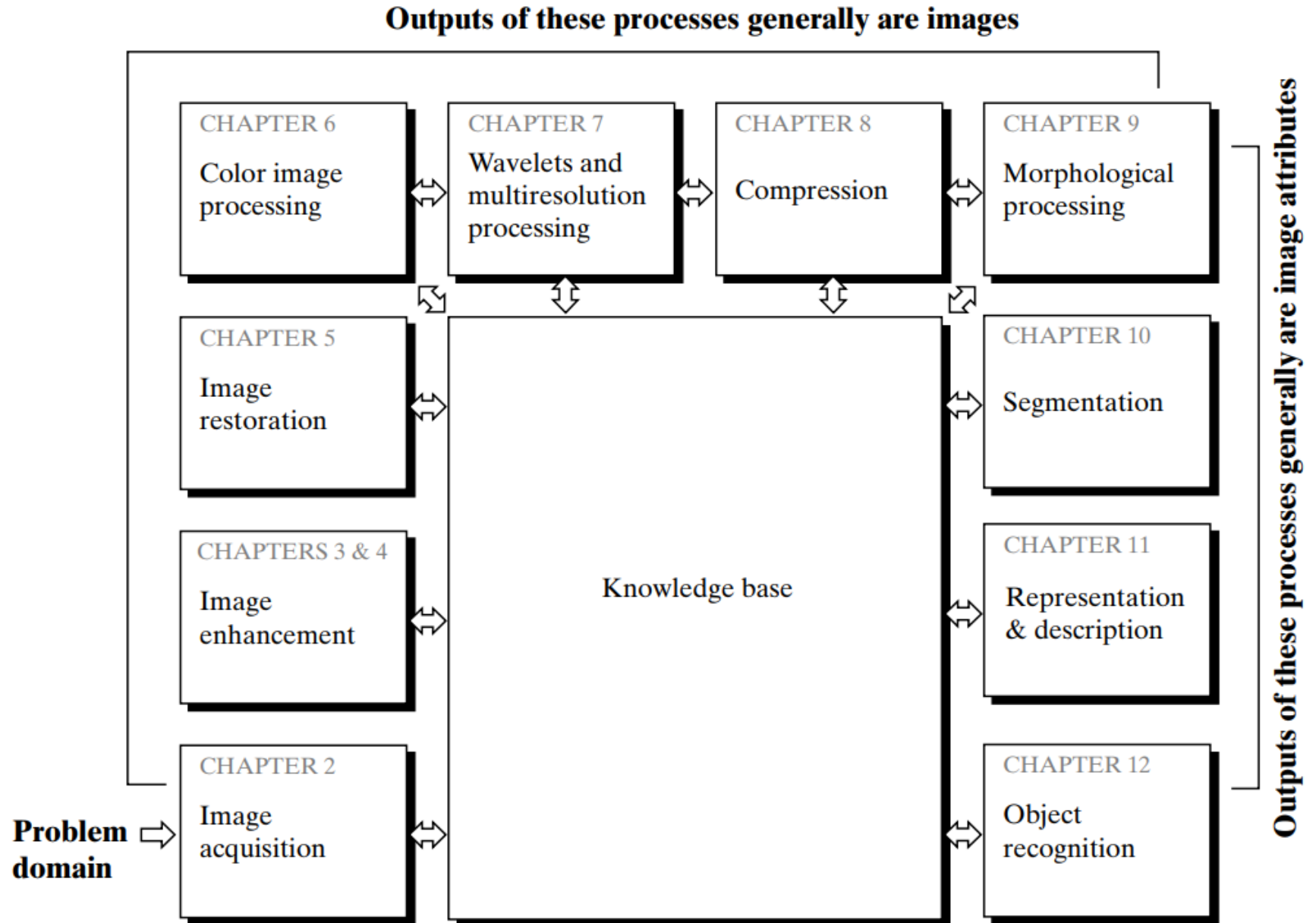
2.9 Ultrasound Imaging

Ultrasound Imaging, also known as sonography, is a diagnostic technique that uses high-frequency sound waves to create images of the body's internal structures.

It is widely used due to its safety, non-invasive nature, and real-time imaging capability.

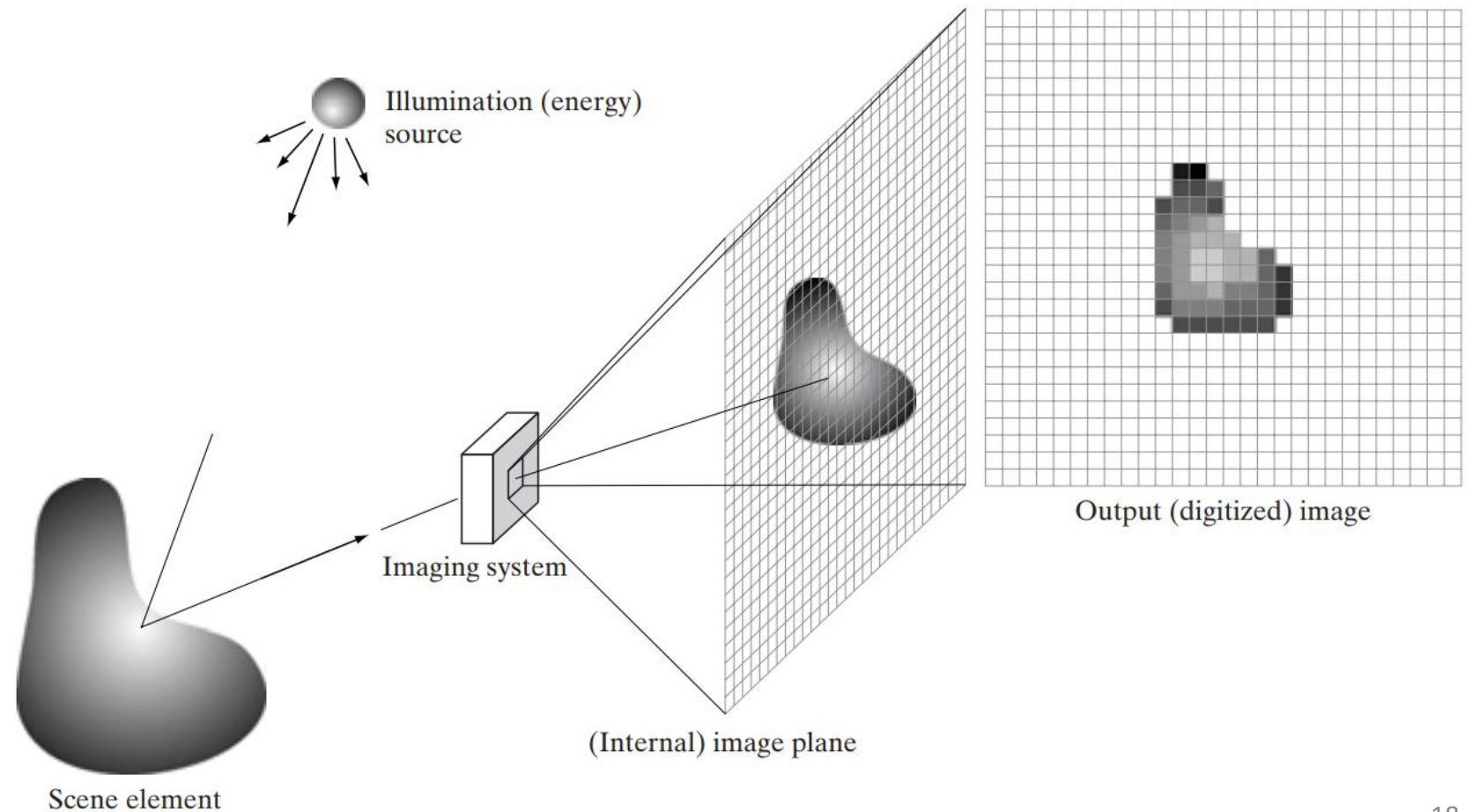


3. Fundamental Steps in Digital Image Processing



3.1 Image Acquisition

Image acquisition refers to the process of *capturing and digitizing visual information from a source* to create a digital image that can be processed, analyzed, or stored.



3.2 Image enhancement

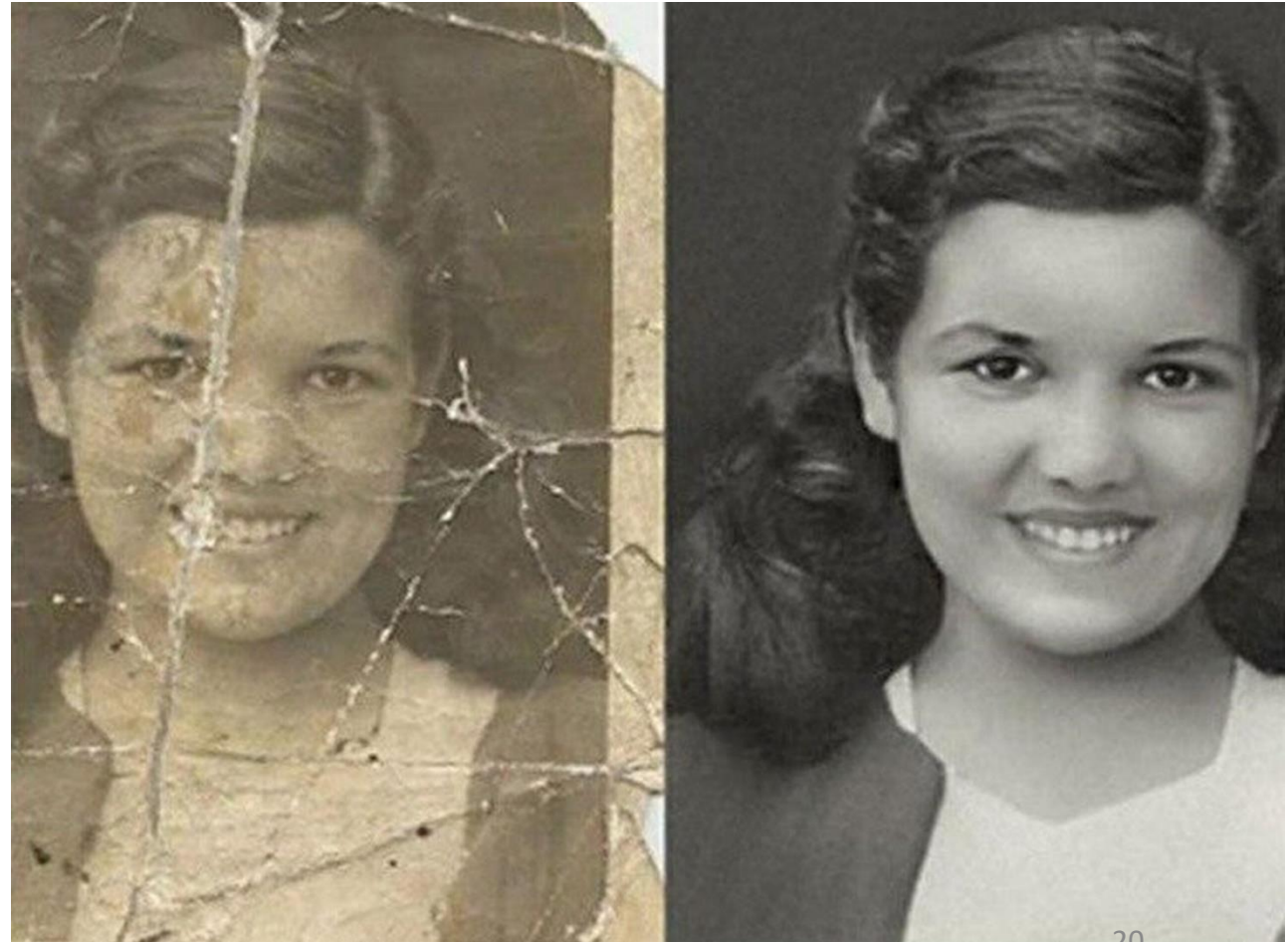
Image enhancement techniques is to bring out detail that is obscured, or simply to highlight certain features of interest in an image.



3.3 Image Restoration

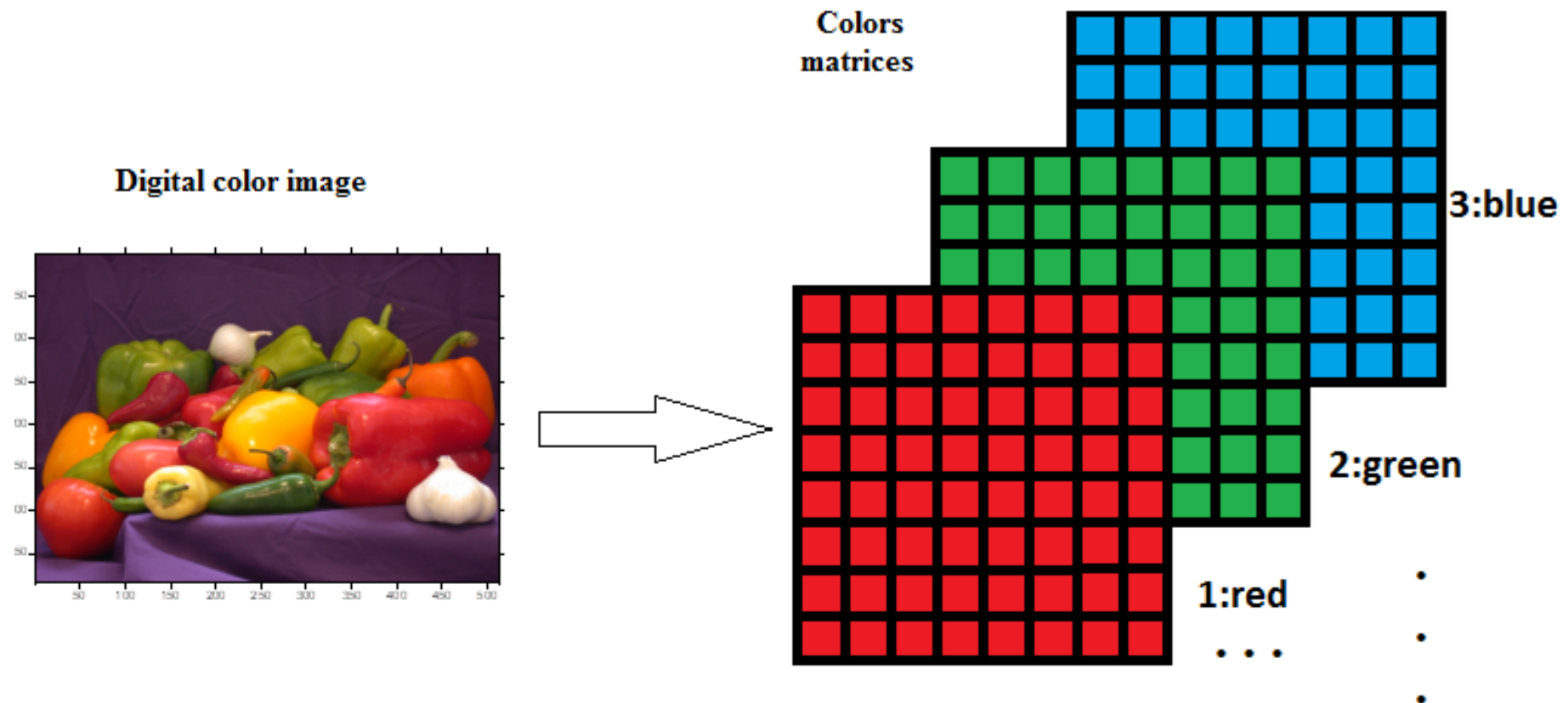
Image restoration is the process of improving the quality of a degraded image by mitigating the effects of distortions or noise.

Unlike image enhancement, which focuses on making an image visually appealing, image restoration aims to reconstruct an image closer to its original, uncorrupted form using mathematical models.



3.4 Color Image Processing

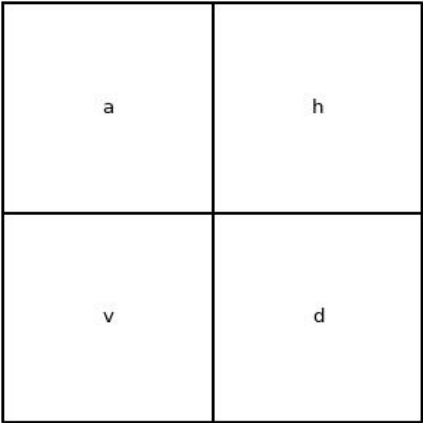
Color Image Processing involves the manipulation, analysis, and enhancement of color images. It utilizes the properties of colors to extract meaningful information, improve image quality, or prepare images for specific applications.



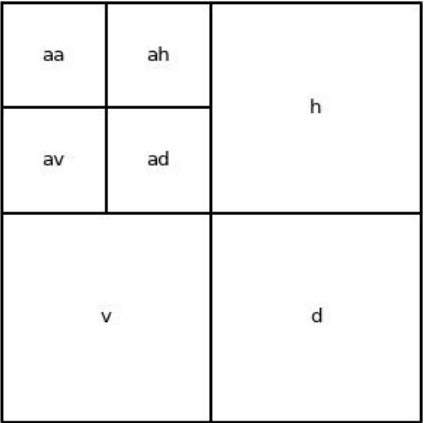
3.5 Wavelet Transform

Wavelets are the foundation for representing images in various degrees of resolution.

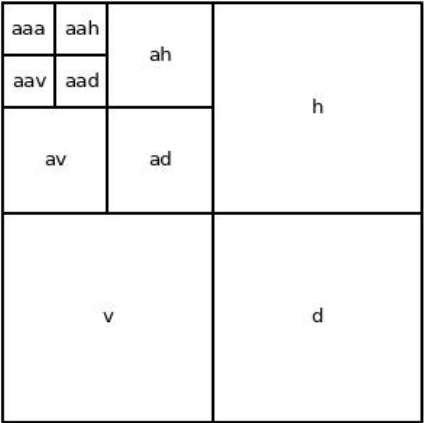
1 level decomposition



2 level decomposition



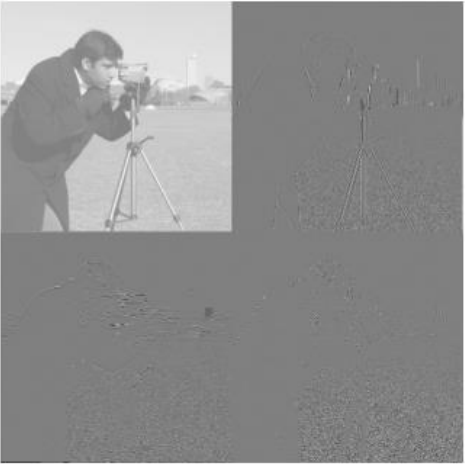
3 level decomposition



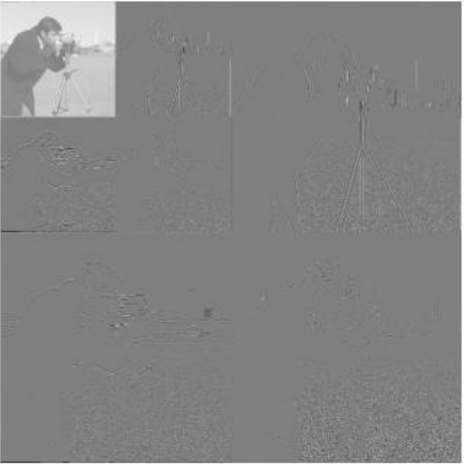
Image



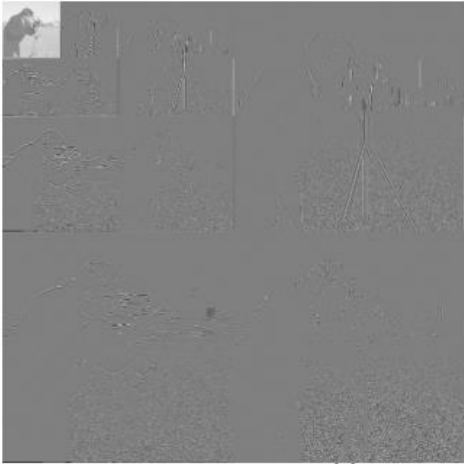
Coefficients (1 level)



Coefficients (2 level)



Coefficients (3 level)



3.6 Compression

Compression, as the name implies, deals with techniques for reducing the storage required to save an image, or the bandwidth required to transmit it.

Original



5.7 MB

Compressed



470 KB

3.7 Morphological Processing

Morphological Processing is a set of image processing techniques that focus on the shape or structure of objects within an image.

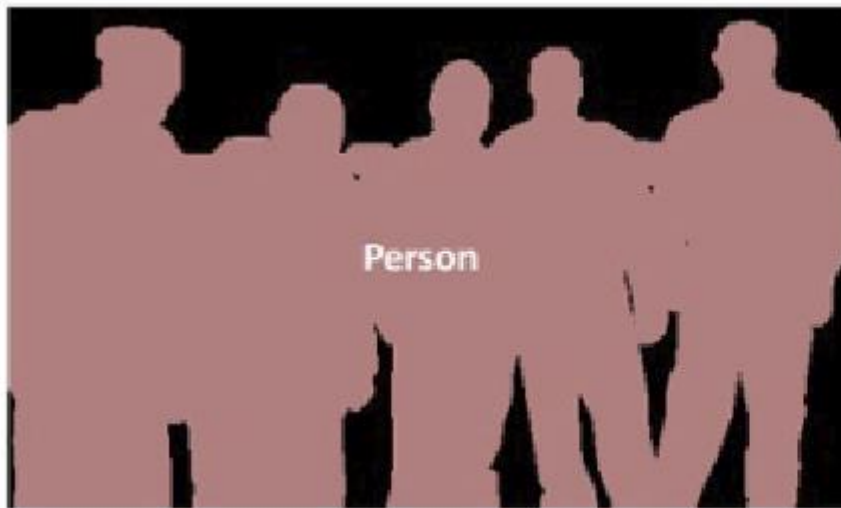
It is particularly useful in binary (black-and-white) images, where the goal is to process and extract meaningful features based on the geometric properties of the objects.



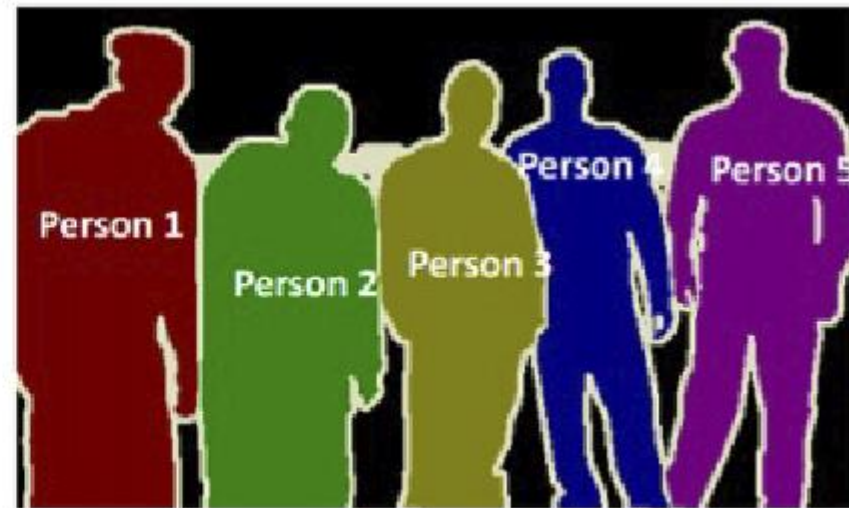
3.8 Segmentation

Segmentation in image processing refers to the process of dividing an image into multiple segments or regions to simplify its representation and make it easier to analyze.

Each segment (or region) typically represents a meaningful part of the image, such as an object, boundary, or texture.



Semantic Segmentation

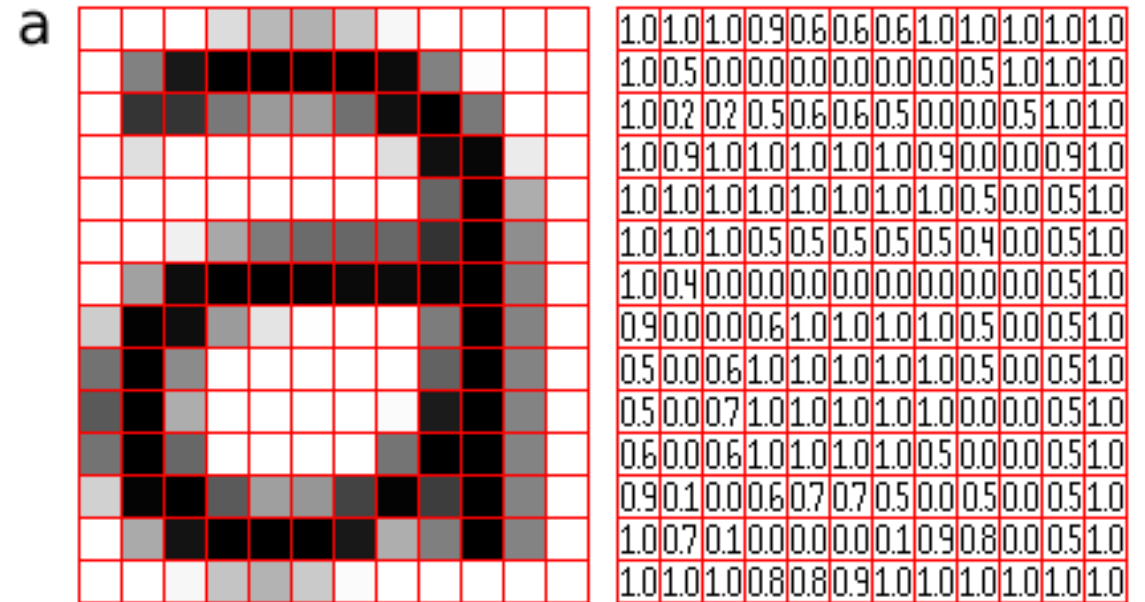


Instance Segmentation

3.9 Representation and Description

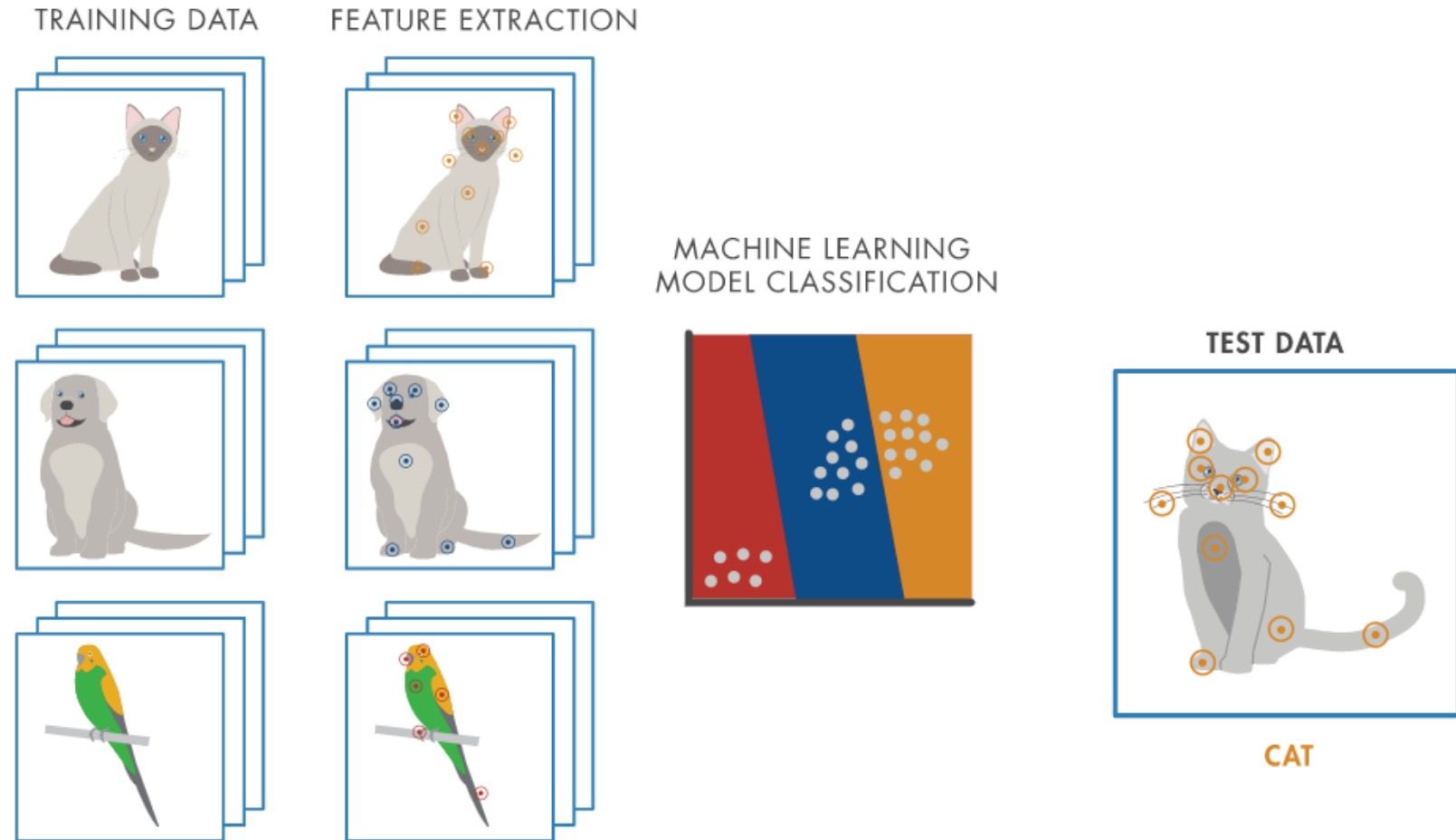
Representation and Description are two fundamental concepts in image processing and computer vision that focus on how to capture and interpret the information in an image after segmentation.

These processes help transform the segmented image into a form that can be easily analyzed and used for further tasks like classification, recognition, or decision-making.



3.10 Recognition

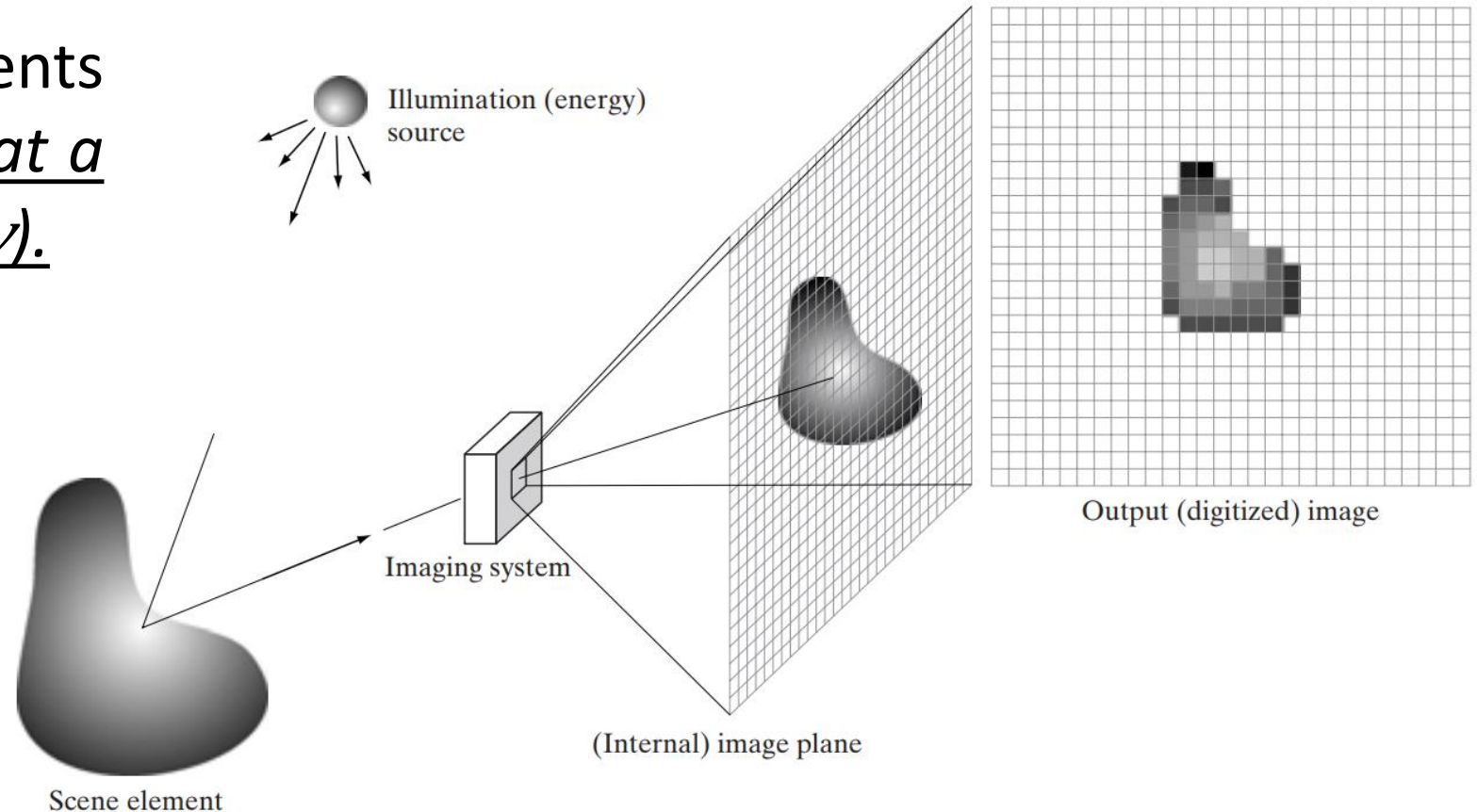
Recognition is the process that assigns a label (e.g., “vehicle”) to an object based on its descriptors.



4. A Simple Image Formation Model

When an image is captured, its pixel values represent the energy emitted or reflected by a physical source (e.g., light, infrared, X-rays, etc.). This energy is detected and quantified to form the image.

The function $f(x,y)$ represents the intensity of the image at a specific pixel located at (x,y) .



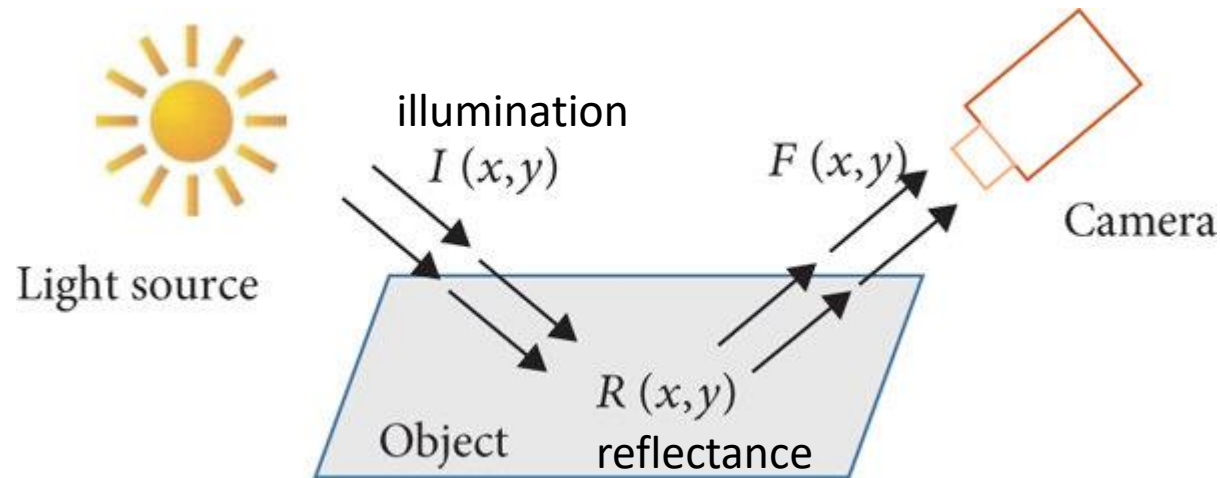
As a consequence, $f(x, y)$ must be nonzero and finite; that is,

$$0 < f(x, y) < \infty.$$

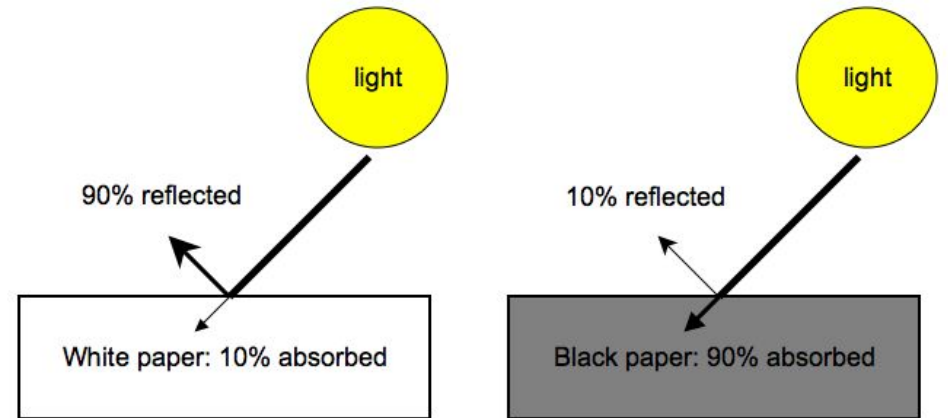
The function $f(x, y)$ represents the intensity of an image at a specific point (x, y) .

This intensity is determined by two components: illumination and reflectance

$$f(x, y) = i(x, y)r(x, y)$$



Surface reflectance



4.1 Illumination Component $i(x,y)$

Represents the amount of light (or energy) falling onto the scene at the point (x,y) .

$$0 < i(x, y) < \infty$$

It depends on the light source and how much light it provides to the scene.

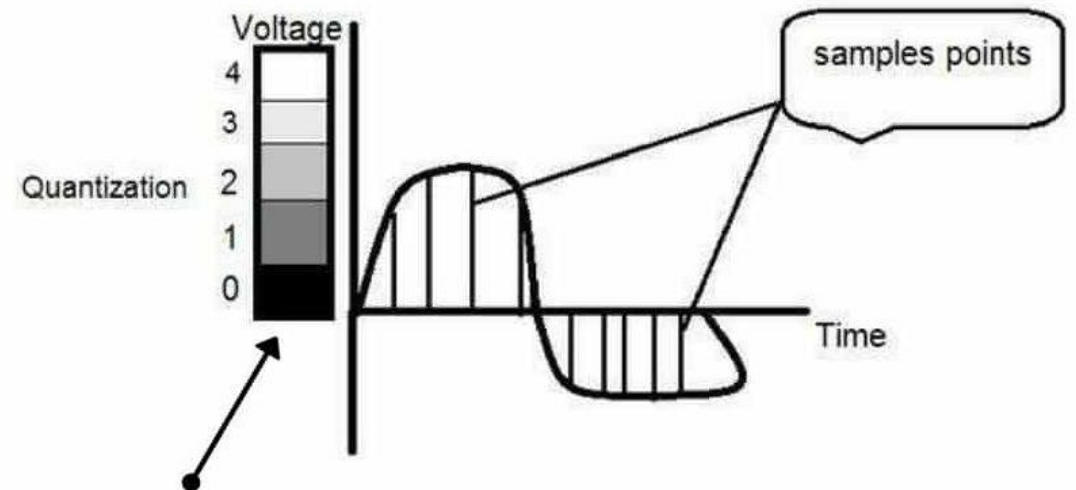
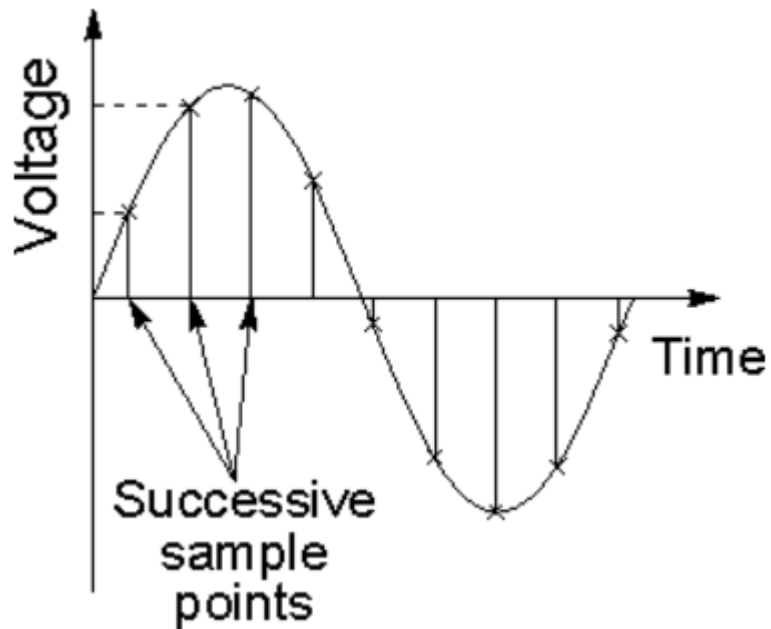
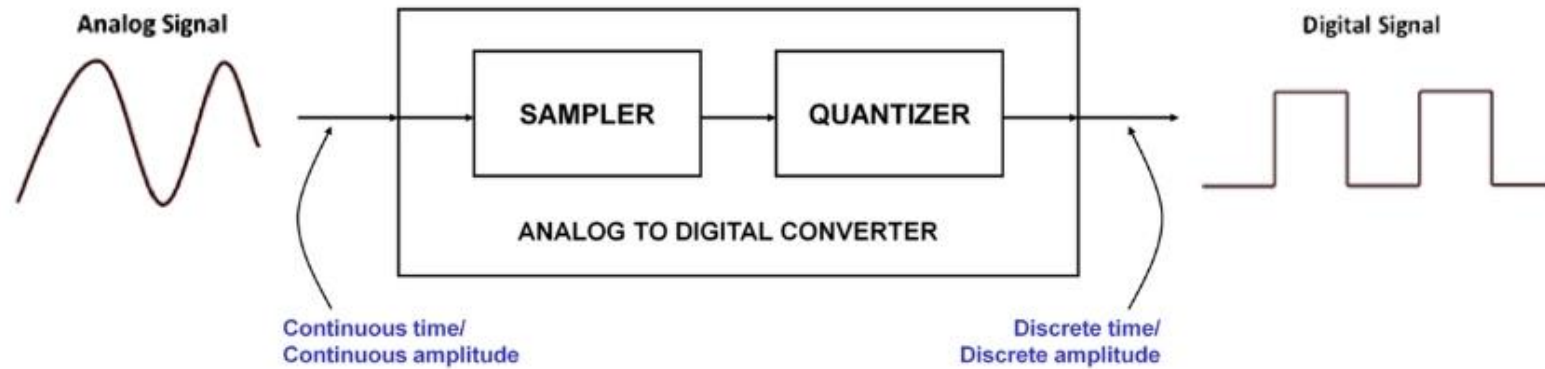
4.2 Reflectance Component $r(x,y)$

Represents the proportion of the incident light that is reflected by objects in the scene at the point (x,y) .

$$0 < r(x, y) < 1$$

Its values range between 0 (no reflection, perfect absorber) and 1 (perfect reflection).

5. Image Sampling and Quantization

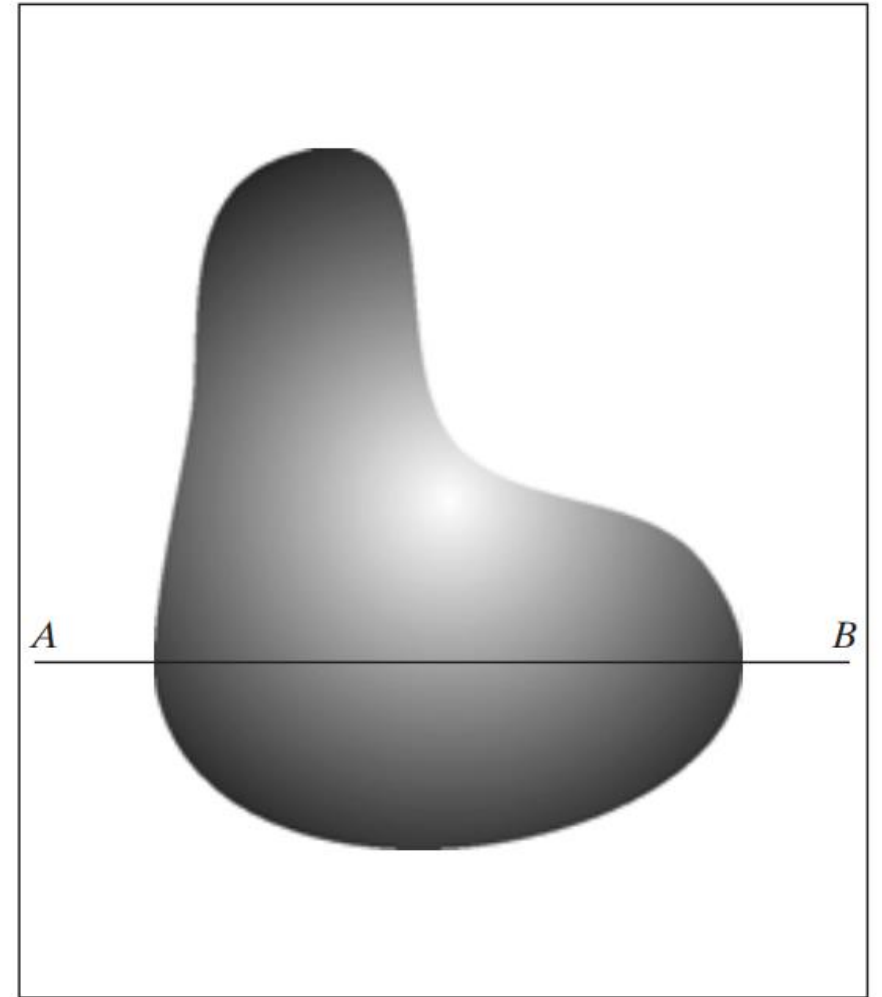


Vertically ranging values have been quantized into 5 different levels or partitions.

To create a digital image, we need to convert the continuous sensed data into digital form.

This involves two processes: sampling and quantization.

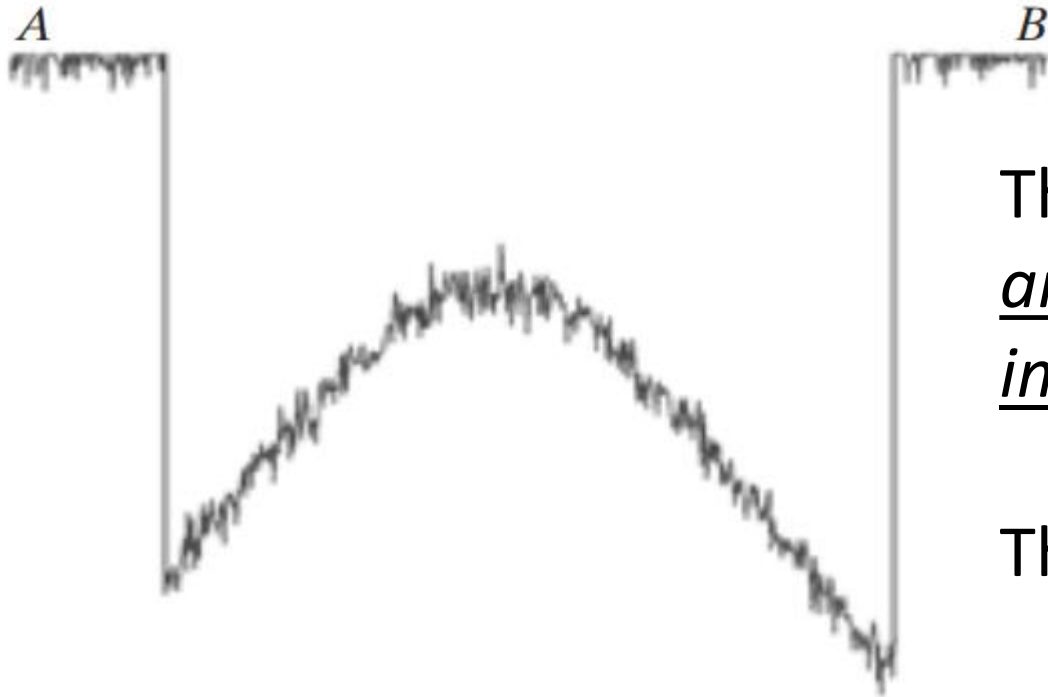
An image may be continuous with respect to the x- and y-coordinates, and also in amplitude.



To convert it to digital form, we have to sample the function in both coordinates and in amplitude.

❑ Digitizing the coordinate values is called sampling.

❑ Digitizing the amplitude values is called quantization.



The one-dimensional function is a plot of amplitude (gray level) values of the continuous image along the line segment AB.

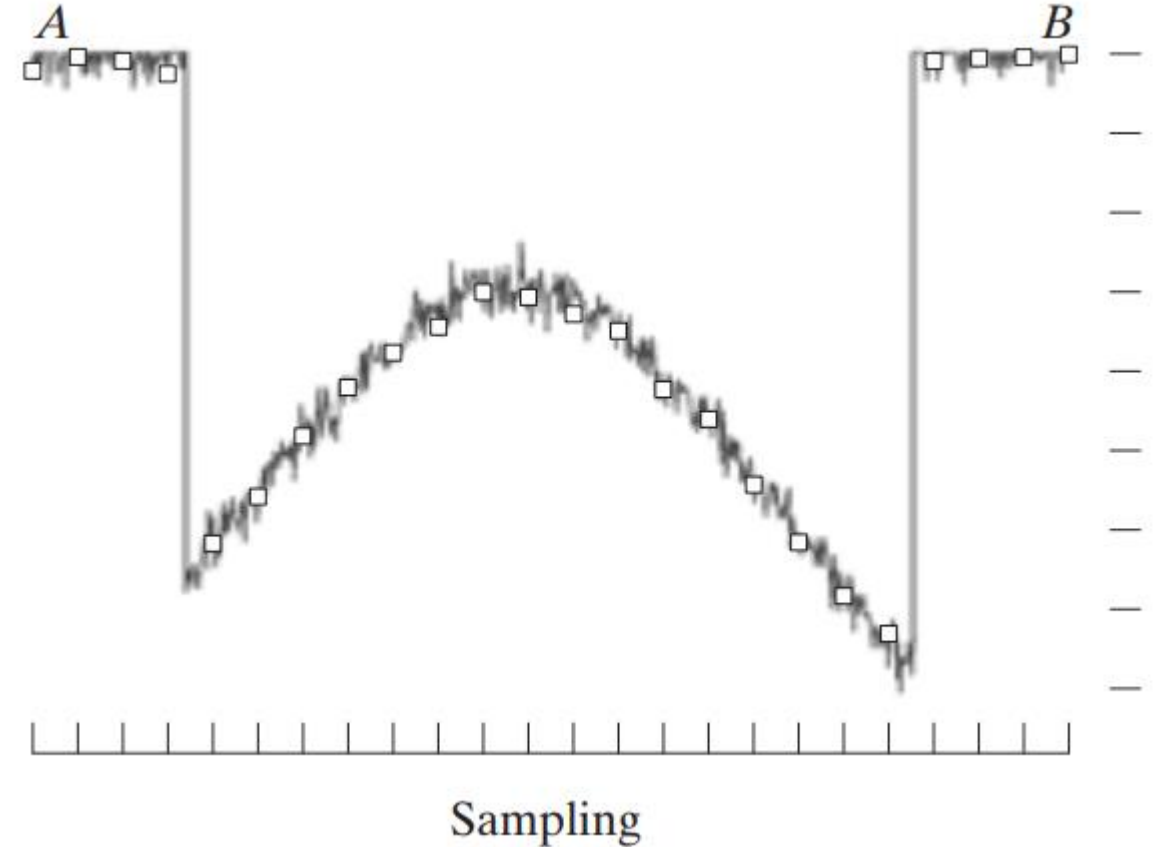
The random variations are due to image noise.

5.1 Sampling

To sample this function, we take equally spaced samples along line AB

The location of each sample is given by a vertical tick mark in the bottom part of the figure.

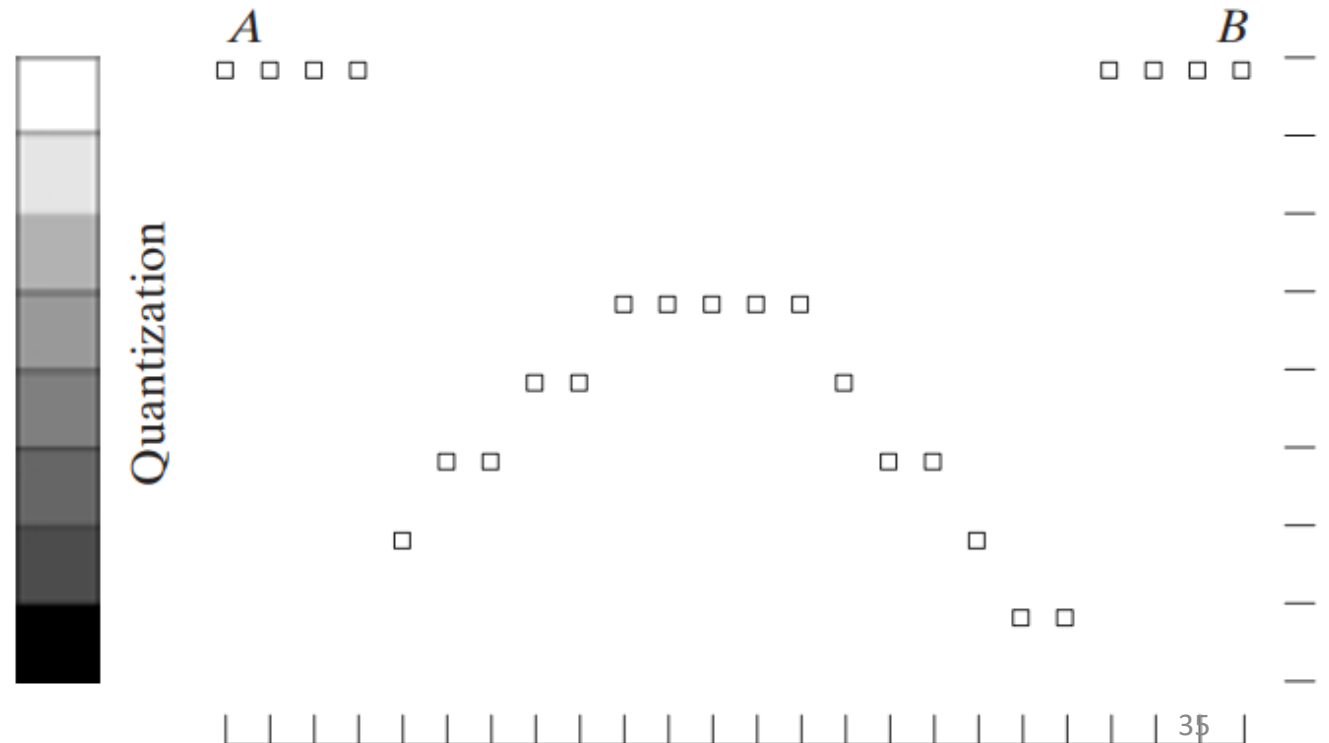
The samples are shown as small white squares superimposed on the function.



5.2 Quantization

The horizontal tick marks indicate the specific value assigned to each of the eight gray levels.

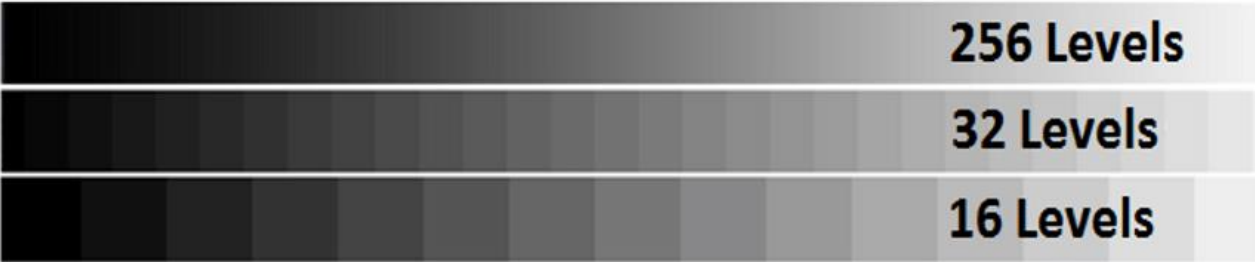
The continuous gray levels are quantized simply by assigning one of the eight discrete gray levels to each sample.



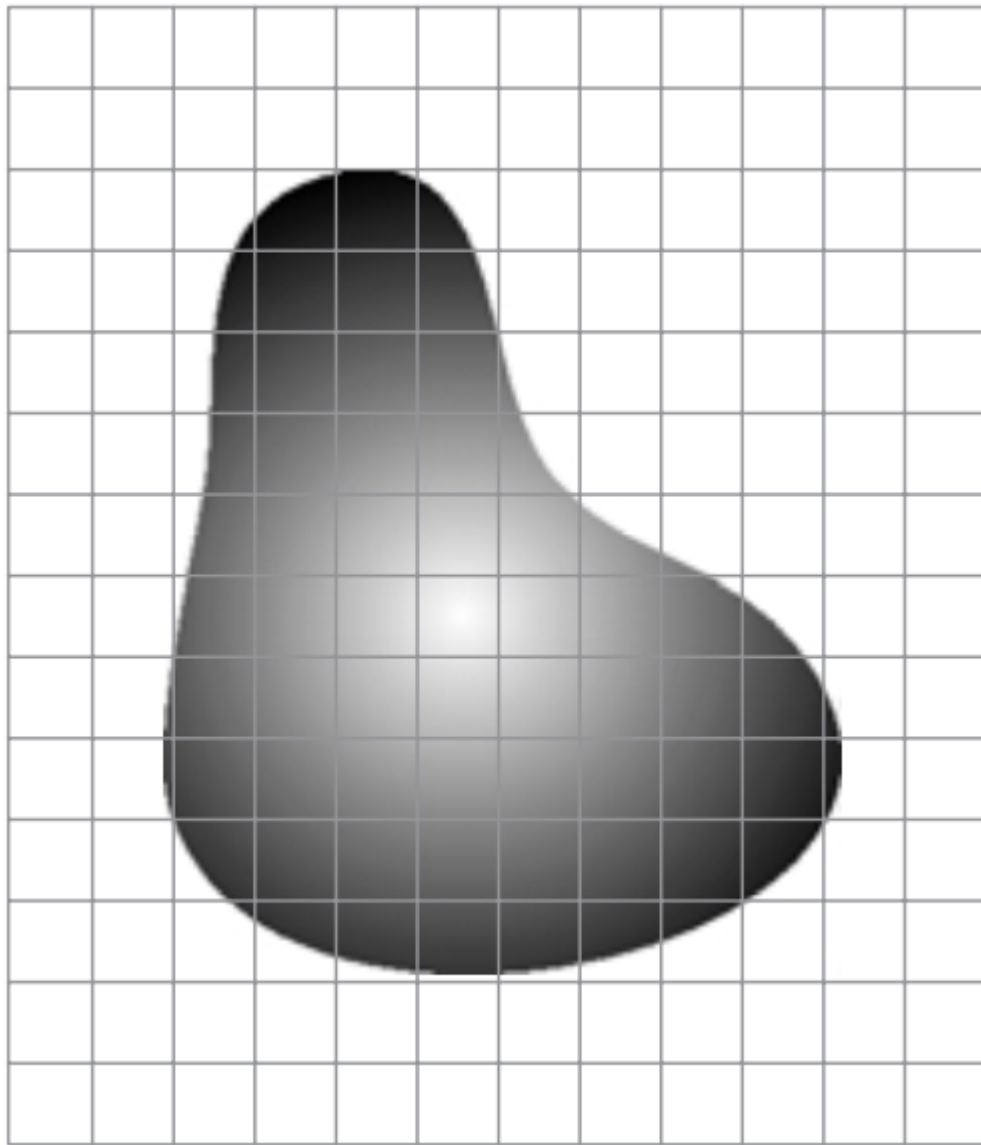
5.2.1 Gray Level

Gray level in an image refers to the intensity or brightness of a pixel in a grayscale image.

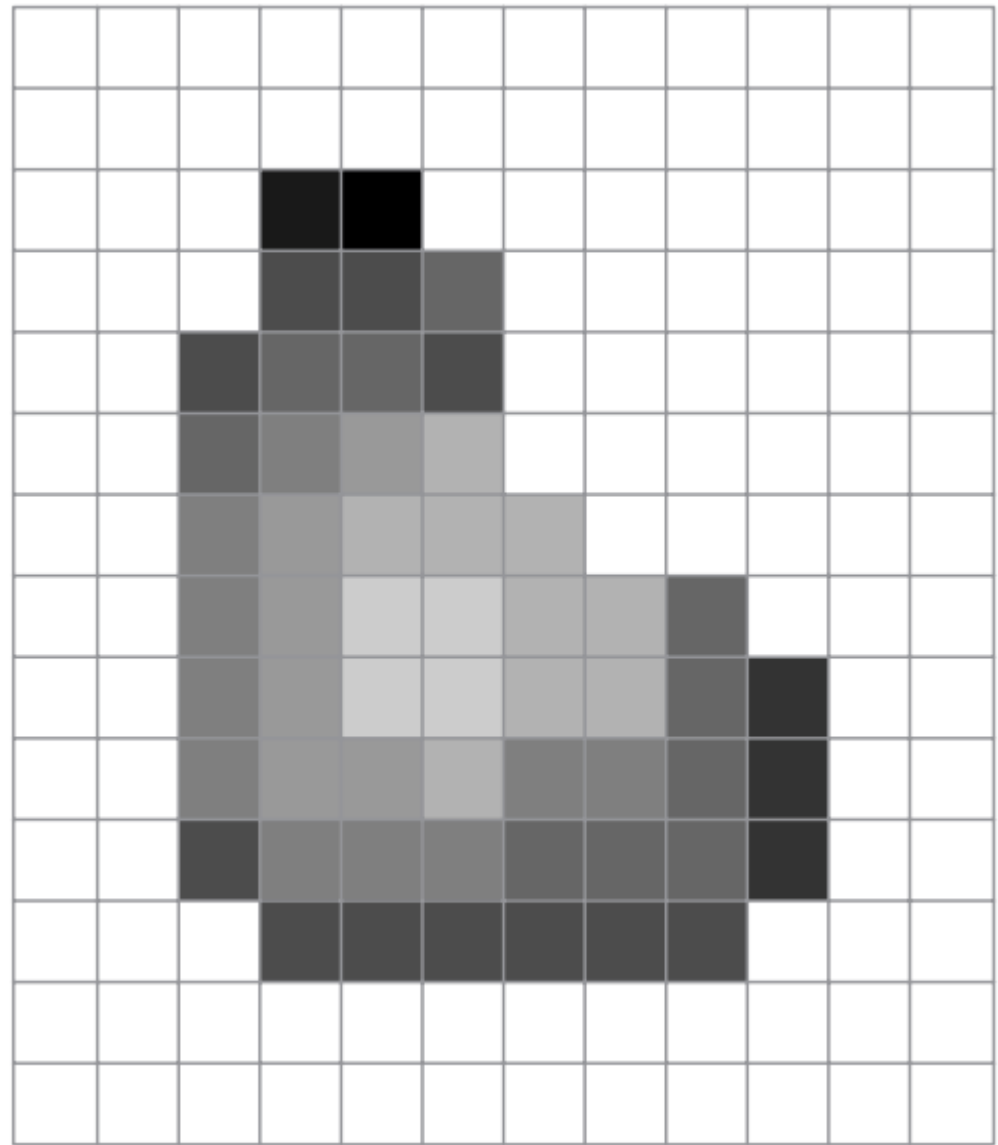
It represents the amount of light at each pixel, ranging from black to white, with varying shades of gray in between.



with getScaledInstance()	gray	without getScaledInstance()
	0	
	16	
	32	
	48	
	64	
	80	
	96	
	112	
	128	
	144	
	160	
	176	
	192	
	208	
	224	
	240	
	255	



Continuous image projected
onto a sensor array



Result of image sampling and
quantization

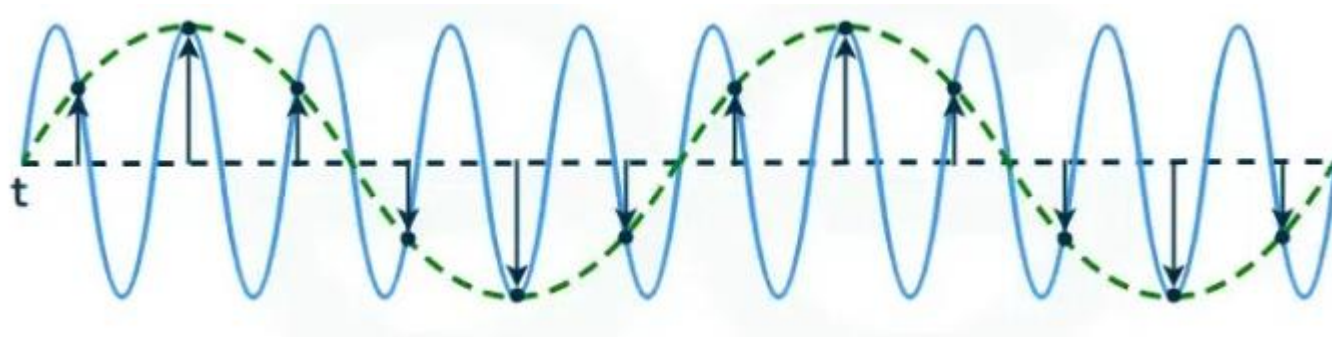
5.3 Issues from Sampling

Low Sampling Rate

Results in aliasing, where the image appears blocky or loses detail because the resolution is too low.

Aliasing Effect

When a continuous signal is sampled at a frequency that is too low to accurately represent the original signal.



5.3.1 How to Improve

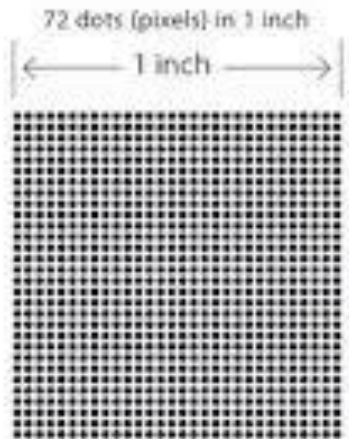
Increase Sampling Resolution:

- ❑ Use more samples (pixels) per unit area to capture finer details.
- ❑ Use a higher DPI (dots per inch) for scanning or image acquisition.

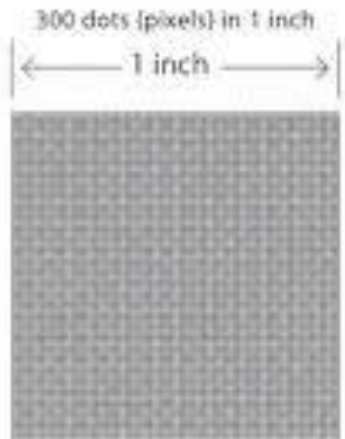
DPI in Images (Printing)

DPI measures the number of dots of ink a printer lays down in an inch.

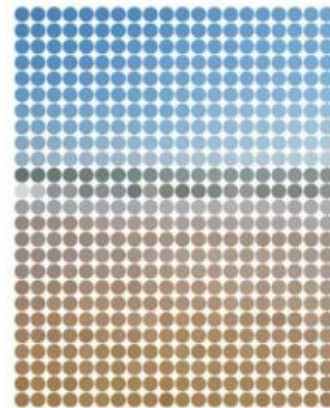
Higher DPI means the printer uses more dots per inch, producing finer and more detailed prints.



72 dpi
72 dots per-inch



300 dpi
300 dots per-inch

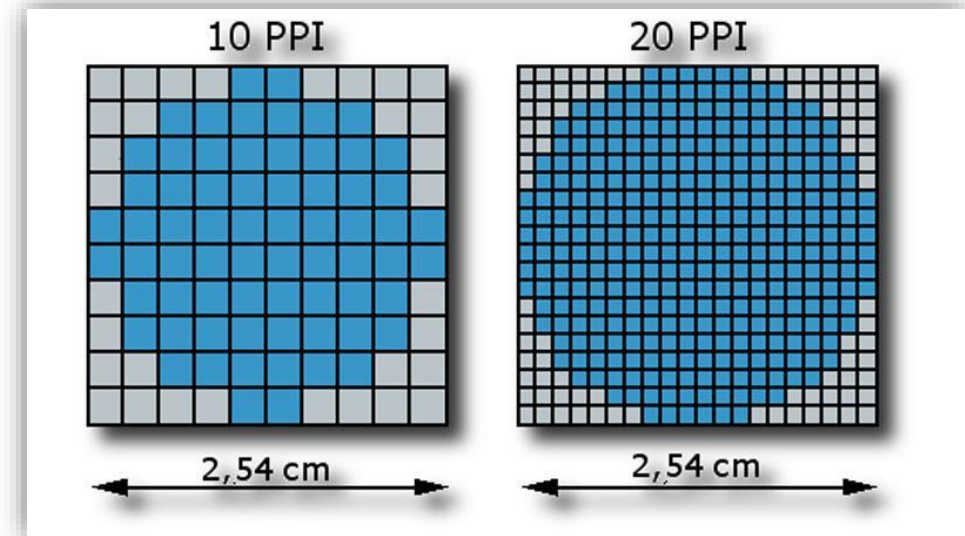
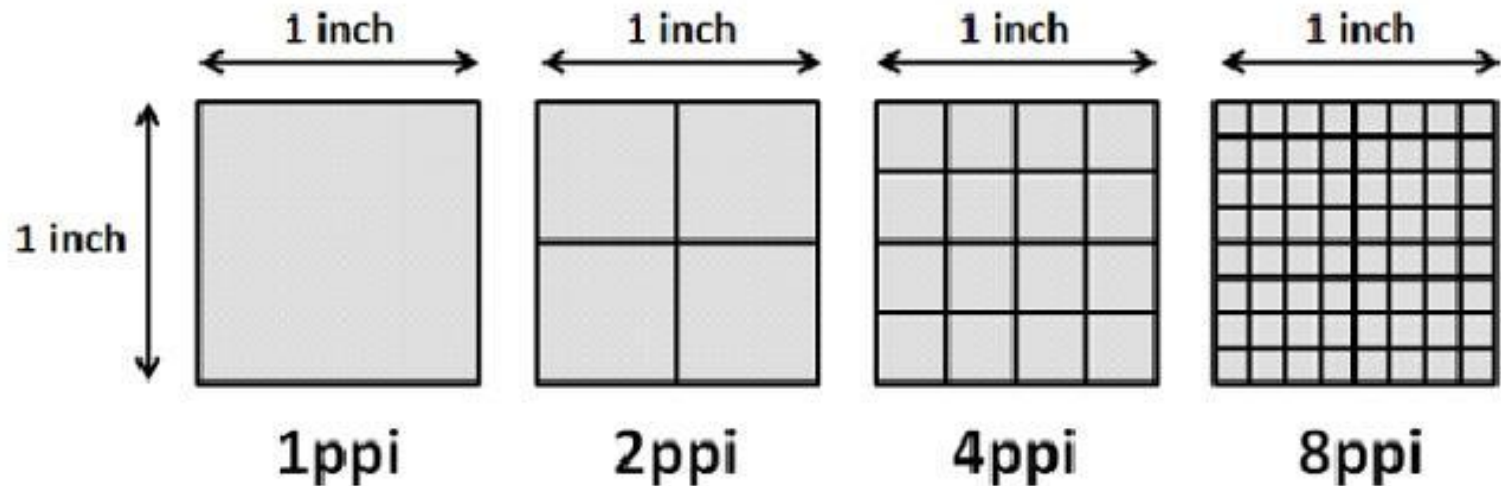


Increasing Dots Per Inch
= More Detailed Image

PPI in Images (Digital Displays)

PPI measures how many pixels are packed into an inch of a digital display or image.

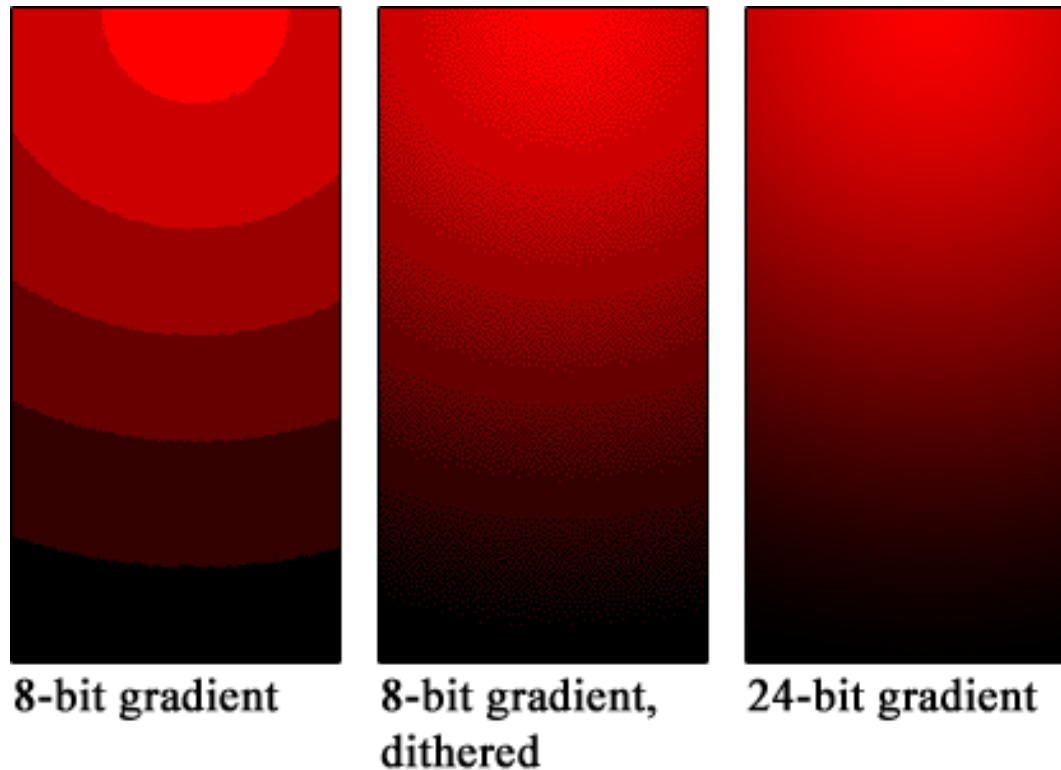
A higher PPI means more pixels per inch, resulting in sharper, clearer images on screens.



5.4 Issues from Quantization

Low Gray-Level Resolution: Fewer intensity levels can cause banding artifacts or a cartoon-like appearance in smooth gradients.

Loss of Fine Intensity Details: Subtle variations in intensity are lost.



5.4.1 How to Improve

Increase Bit Depth:

Use more bits per pixel to allow more intensity levels. Switch from 8-bit (256 levels) to 16-bit (65,536 levels) for medical or scientific imaging.

Dithering:

Add controlled noise to the image before quantization to reduce visible artifacts and make transitions smoother.



8-bit gradient



8-bit gradient,
dithered



24-bit gradient